

MONITORING REPORT FOR LUANGWA COMMUNITY FORESTS PROJECT

CCB-001

Document Prepared by BioCarbon Partners

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GHG Accounting Period	01-May-2015 to 30-April-2045; 30-year lifetime
Monitoring Period of this Report	01-May-2015 to 31-December-2022
History of CCB Status	CCB Validation: 2019
Gold Level Criteria	Community – improved livelihood and well-being of low income and vulnerable communities Climate – climate change adaptation through the diversification of income streams, resilient farming and the preservation of habitat corridors Biodiversity – improved protection of multiple threatened and endemic species

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1 SUMMARY OF PROJECT BENEFITS

1.1 Unique Project Benefits

Table 1. Unique Project Benefits

Outcome or Impact	Achievements during the Monitoring Period (2015-2022)	Section Reference	Achievements during the Project Lifetime (2015-2022)
1) Conserving a large part of the Luangwa catchment, one of the longest undammed rivers in Southern Africa	Through conservation activities, enhancement of law enforcement measures and poverty alleviation, the project has been effective in managing 1,236,257 ha of the Luangwa river catchment area	4 5	Through conservation activities, enhancement of law enforcement measures and poverty alleviation, the project has been effective in managing 1,236,257 ha of the Luangwa river catchment area
2) Conserving the world's largest hippopotamus population	The project has successfully improved the protection of almost 326 km (40%) of the Luangwa river, which has simultaneously improved the conservation of the hippopotamus population. In 2011 it was estimated that 25,000 individual hippopotamus were in the whole Luangwa river ¹ . The regular hippopotamus counts conducted by the Department of National Park and Wildlife over more than 200 km or River stretch shows that the population remained stable since the start of the project (Howard Maimbo, personal communication).	5	The project has successfully improved the protection of almost 326 km (40%) of the Luangwa river, which has simultaneously improved the conservation of the hippopotamus population. In 2011 it was estimated that 25,000 individual hippopotamus were in the whole Luangwa river ¹ . The regular hippopotamus counts conducted by the Department of National Park and Wildlife over more than 200 km or River stretch shows that the population remained stable since the start of the project (Howard Maimbo, personal communication).
3) Conserving a near continuous wildlife corridor between five national parks	The project strengthens the sustainable management of over 1 million ha of wildlife habitat (Game management Areas and Private Game Ranches), therefore improving the connectivity between the five National Parks bordering the project area. The importance of maintaining the ecological function of this network	3 5	The project strengthens the sustainable management of over 1 million ha of wildlife habitat (Game management Areas and Private Game Ranches), therefore improving the connectivity between the five National Parks bordering the project area. The importance of maintaining the ecological function of this network

¹ Wilbrod, C. and Milanzi, J., 2011. Population status of the hippopotamus in Zambia. *African Journal of Ecology*, 49(1), pp.130-132.

Outcome or Impact	Achievements during the Monitoring Period (2015-2022)	Section Reference	Achievements during the Project Lifetime (2015-2022)
	of Protected Areas has been illustrated a few times through long distance dispersal of African wild dogs during the current monitoring period ² .		of Protected Areas has been illustrated a few times through long distance dispersal of African wild dogs during the current monitoring period.
4) Benefiting over 100,000 local community members	The project has been beneficial to over 222,894 community members across 17 Chiefdoms	4	The project has been beneficial to over 222,894 community members across 17 Chiefdoms
5) Carbon Sales, Community Revenue, Shares	Since 2019 more than \$ 10 million cumulative payments have been made directly to the communities within the project areas and subsequently enhancing household livelihood	4	Since 2019 more than \$ 10 million cumulative payments have been made directly to the communities within the project areas and subsequently enhancing household livelihood

1.2 Standardized Benefit Metrics

Table 2. Standardized Benefit Metrics

Category	Metric	Achievements during Monitoring Period (2015-2022)	Section Reference	Achievements during the Project Lifetime (2015-2022)
GHG emission reductions &	Net estimated emission removal in the project area, measured against the without-project scenario	N/A	N/A	N/A
	Net estimated emission reductions in the project area, measured against the without-project scenario	11,298,456 tCO ₂ e	3.1.1	11,298,456 tCO ₂ e
Forest ³ cover	REDD ⁴ projects: the number of hectares of reduced forest loss within the project area are measured against the without-project scenario	47,835	3.1.1	47,835

² Zambian Carnivore Programme. Annual Report 2021.

³ Land with woody vegetation that meets an internationally accepted definition (e.g., UNFCCC, FAO or IPCC) of what constitutes a forest, which includes threshold parameters, such as minimum forest area, tree height and level of crown cover, and may include mature, secondary, degraded and wetland forests (*VCS Program Definitions*)

⁴ Reduced emissions from deforestation and forest degradation (REDD) - Activities that reduce GHG emissions by slowing or stopping conversion of forests to non-forest land and/or reduce the degradation of forest land where forest biomass is lost (*VCS Program Definitions*)

	ARR ⁵ projects: the number of hectares of forest cover increased in the project area measured against the without-project scenario	N/A	N/A	N/A
Improved land management	Number of hectares of existing production forest land in which IFM ⁶ practices have occurred as a result of the project's activities, measured against the without-project scenario	N/A	N/A	N/A
	Number of hectares of non-forest land in which improved land management has occurred as a result of the project's activities, measured against the without-project scenario	408.5 ha	4.1 4.3	408.5 ha
Training	Total number of community members who have improved skills and/or knowledge resulting from training provided as part of project activities	1783 community members	2.2.1 4 4.3.1	1783 community members
	Number of female community members who have improved skills and/or knowledge resulting from training provided as part of project activities	688	2.2.1 4 4.3.1	688
Employment	Total number of people employed in project activities ⁷ expressed as number of full time employees ⁸	Out of 242 staff members in BCP 136 are full time employees in LCFP (as of 30/11/2022)	2 4	Out of 242 staff members in BCP 136 are full time employees in LCFP (as of 30/11/2022)
	Number of women employed in project activities, expressed as number of full time employees	33 BCP employees including 23 community scouts (as of 30/11/2022)	4	33 BCP employees including 23 community scouts (as of 30/11/2022)

⁵ Afforestation, reforestation and revegetation (ARR) - Activities that increase carbon stocks in woody biomass (and in some cases soils) by establishing, increasing and/or restoring vegetative cover through the planting, sowing and/or human-assisted natural regeneration of woody vegetation (*VCS Program Definitions*)

⁶ Improved forest management (IFM) - Activities that change forest management practices and increase carbon stock on forest lands managed for wood products such as saw timber, pulpwood and fuelwood (*VCS Program Definitions*)

⁷ Employed in project activities are people directly working on project activities in return for compensation (financial or otherwise), including employees, contracted workers, sub-contracted workers and community members that are paid to carry out project-related work.

⁸ Full time equivalency is calculated as the total number of hours worked (by full-time, part-time, temporary and/or seasonal staff) divided by the average number of hours worked in full-time jobs within the country, region or economic territory (adapted from UN System of National Accounts (1993) paragraphs 17.14[15.102]; [17.28])

Livelihoods	Total number of people with improved livelihoods ⁹ and/or income generated as a result of project activities	28,631	4.3.1	28,631
	Number of women with improved livelihoods or income generated as a result of project activities	16,579	4.3.1	16,579
Health	Total number of people for whom health services were improved as a result of project activities, measured against the without-project scenario	17,712	4.3.1	17,712
	Number of women for whom health services were improved as a result of project activities, measured against the without-project scenario	9,111	4.3.1	9,111
Education	Total number of people for whom access to, or quality of, education was improved as a result of project activities, measured against the without-project scenario	36,244	4.3.1	36,244
	Number of women and girls for whom access to, or quality of, education was improved as a result of project activities, measured against the without-project scenario	17,191	4.3.1	17,191
Water	Total number of people who experienced increased water quality and/or improved access to drinking water as a result of project activities, measured against the without-project scenario	26,915	4.3.1	26,915
	Number of women who experienced increased water quality and/or improved access to drinking water as a result of project activities, measured against the without-project scenario	13,968	4.3.1	13,968

⁹ Livelihoods are the capabilities, assets (including material and social resources) and activities required for a means of living (Krantz, Lasse, 2001. *The Sustainable Livelihood Approach to Poverty Reduction*. SIDA). Livelihood benefits may include benefits reported in the Employment metrics of this table.

Well-being	Total number of community members whose well-being ¹⁰ was improved as a result of project activities	Overall, 120,386 project benefits have been recognized by the community members from both genders, shared within 11 categories. On average, each category benefits over 10,944 community members (range: 541-36244) out of a total of 222,894 people.	4.1.3 4.3.1	Overall, 120,386 project benefits have been recognized by the community members from both genders, shared within 11 categories. On average, each category benefits over 10,944 community members (range: 541-36244) out of a total of 222,894 people.
	Number of women whose well-being was improved as a result of project activities	Overall, 60,302 project benefits have been recognized by the female community members, shared within 11 categories. On average, each category benefits over 3,142 female community members (range: 271-17,191) out of a total of 114,757 women.	4.3.1 4.4.3	Overall, 60,302 project benefits have been recognized by the female community members, shared within 11 categories. On average, each category benefits over 3,142 female community members (range: 271-17,191) out of a total of 114,757 women.
Biodiversity conservation	Change in the number of hectares significantly better managed by the project for biodiversity conservation ¹¹ , measured against the without-project scenario	1,236,257 ha	3.3.1 5.1.1 5.2.1 5.3.1	1,236,257 ha

¹⁰ Well-being is people's experience of the quality of their lives. Well-being benefits may include benefits reported in other metrics of this table (e.g. Training, Employment, Health, Education, Water, etc.), but could also include other benefits such as empowerment of community groups, strengthened legal rights to resources, conservation of access to areas of cultural significance, etc.

¹¹ Biodiversity conservation in this context means areas where specific management measures are being implemented as a part of project activities with an objective of enhancing biodiversity conservation.

Biodiversity conservation	Number of globally Critically Endangered or Endangered species ¹² benefiting from reduced threats as a result of project activities ¹³ , measured against the without-project scenario	Critically Endangered: 3 Endangered: 4	2.1.3 5.1.1 5.1.4 5.3.1 5.4.1	Critically Endangered: 3 Endangered: 4
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¹² Per IUCN's Red List of Threatened Species

¹³ In the absence of direct population or occupancy measures, measurement of reduced threats may be used as evidence of benefit

2 GENERAL

2.1 Project Goals, Design and Long-Term Viability

2.1.1 Implementation Schedule

Table 3. Implementation Schedule

Date	Milestone(s) in the project's development and implementation
2014-02-01	BCP Awarded USAID Grant
2015-05-01	Beginning of forest demarcation by chiefdoms. Project start date and beginning of first monitoring (crediting) period
2016-08-08	Department of National Parks and Wildlife approval
2016-11-01	Infrastructure Development starts
2016-12-01	Beekeeping starts
2017-05-01	Conservation farming starts
2017-07-25	Forestry Department Approval
2017-08-01	CRB-BCP Forest Monitoring starts
2018-12-31	End of first VCS monitoring period running from project start date. Thereafter monitoring periods will be annual for VCS and follow the calendar year (e.g. 2019-01-01 to 2019-12-31) except for the last monitoring period which may run from 2044-01-01 to 2045-05-01
2019-04-16	Approval of the first 7 project area CFMAs by Director of Forestry
2019-9-30	Expected first CCB-VCS validation and verification date.
2019-10-14	Actual first CCB-VCS validation and verification by SCS Global Services
2020-06-20	Second VCS verification by AENOR
2020-12-31	Planned end of first CCB monitoring period as presented in the PD
2021-04-19	Third VCS verification by AENOR
2022-11-09	Fourth VCS verification by AENOR
2023-02-07	Planned submission of the first CCB MR (monitoring period 2015-05-01 to 2022-12-31) and fifth VCS verification (monitoring period 2022-01-01 to 2022-12 -31)
May 2025	Mandatory baseline re-evaluation as per current VCS AFOLU rules
May 2035	Mandatory baseline re-evaluation as per current VCS AFOLU rules
May 2045	Project crediting period end date

2.1.2 Minor Changes to Project Design (*Rules 3.5.6*)

1) Demonstration of Forest Age

The PDR.22 methodology requires digital GIS-based maps using aerial or satellite imagery of the project accounting areas indicating forests or native grasslands from the project start date as well as 10 years prior.

As this project used PALSAR radar data with the earliest available data from 2007, the requirements were not met. However, the use of PALSAR data was justified as the project areas are within virgin forests that were previously inaccessible with little to no history of large disturbances. Therefore, a 2007 forest map of the areas ought to be indistinguishable from a 2005 map. Additionally, the use of PALSAR imagery results in accurate and detailed maps in comparison to optical imagery used 10 years prior to the start date.

Optical imagery is best suited for mapping during the months of April to June as this time period is relatively cloud-free and trees within the forests have leaves. However, just after this period, which is immediately after the rains, results in significant changes in the appearance of the forest in optical imagery. This project initially mapped forests using optical imagery but found significant changes between years that were the result of seasonality rather than real changes in forest areas. These effects were significantly reduced with the use of PALSAR radar data due to its cloud penetrating ability and the L band radar used has a relatively long bandwidth and thus does not generally interact with the leaves of trees, but does interact with large branches and stems¹⁴. Additionally, the VM0009 framework does not require a remote sensing analysis of an area, but rather a visual representation at the start date as well as 10 years prior. Such an approach is highly subjective as unconverted grasslands and farmlands may at times appear similar. With the PALSAR based analysis, all non-forest areas, whether natural or not, were excluded. This improved the accuracy of the analysis and did not negatively impact the conservativeness of the quantification of GHG emission reductions or removals. The deviation only affects the monitoring and measuring of the methodology.

2) Leakage Plot Protocol

A minor deviation of the visual approach for leakage monitoring (Appendix B, VM0009) was integrated. Instead of Forest Inventory Team members estimating degradation, they geo-reference every felled tree in the permanent plots. The total number of felled trees were then compared against the average density of trees/ha in the project accounting area, avoiding biased estimations. This value was then placed in a degradation category as per VM0009. This approach removes all potential bias and greatly improves the robustness and repeatability of degradation monitoring. This deviation does not affect the conservativeness of any emission reductions as it is an improvement of the prescribed procedure and removes observer bias that could result in degradation in the leakage area being underestimated. The deviation only affects the monitoring and measuring section of the methodology.

A section of 19,887 ha of the second project instance areas overlapped with the existing activity, shifting the leakage area which was originally 319,179 ha, to the new leakage area comprising 299,291 ha. This is not strictly regarded as a methodological deviation, as VM0009 allows for existing leakage areas to be converted to future project activity instance areas.

3) Biomass estimation

The project uses two different allometric equations to estimate above-ground biomass. The first equation was developed using data provided by Professor Chudimayo from Zambian Miombo (Appendix B) who sampled trees of Diameter at Breast Height (DBH) up to 55 cm. The second equation was developed by Mugasha et al. in Tanzanian Miombo (Appendix B) and is used to estimate the biomass of trees between 63 to 110 cm DBH. The biomass of trees with DBH between 55 and 63 cm is assumed to be equal to the

¹⁴ Mitchard, E. T. A., *et al.* "Using satellite radar backscatter to predict above - ground woody biomass: A consistent relationship across four different African landscapes." *Geophysical Research Letters* 36.23 (2009)

biomass of trees with a 55 cm DBH. Likewise, trees with a greater than 110 cm DBH are assumed to have a biomass equal to a 110 cm DBH tree. This parsing is necessary because the methodology requires that the biomass of the largest tree in the representative sample of destructively harvested trees be greater than the biomass predicted by the equation and that the value of the first-order derivative of the largest tree in the representative sample and inventory not differ by more than 10%. The first selected allometric equation under-predicts the biomass of the largest tree in the representative sample, but the difference of the first-order derivative values was greater than 10%. The second allometric equation slightly over predicts the biomass of the largest tree in the sample. However, compared to the Zambian equation, it predicts lower biomass for every tree size. Therefore, the equations were applied as described above to enable the project to make an estimate of the biomass of larger trees that fell outside the range of the size of trees used to develop the equations. It was found that only 33 trees out of 3589 trees (<1%) in the inventory had a DBH above 55 cm and only 3 had a DBH above 110 cm. Applying the equations in this order is conservative and effaces the possibility of the equation over predicting biomass of trees outside the size classes for which it was tested. This deviation only relates to how biomass is measured for monitoring purposes and doesn't affect any other part of the methodology. It results in a more conservative estimate of biomass and emissions savings. Therefore, it meets the criteria for an acceptable methodological deviation.

4) Soil Emission Calculations

The VM0009 baseline soil emission model assumes that soil organic carbon (SOC) in deforested areas will deplete to the same value as the proxy area after 5 years. Project SOC emissions are to be tracked directly through field measurement in encroached areas. However, in the project area, the main drivers of deforestation are smallholder farmers who deliberately choose high fertility areas for conversion with higher than average SOC values. Following the methodology, it would appear that encroachment causes an increase in SOC, which is not consistent with what is actually occurring to soil stocks. To avoid this bias, it is conservatively assumed that once encroached, the average SOC stocking per ha in the encroached area becomes the proxy area SOC stocking. This methodological deviation is, therefore, more conservative in calculating emission reductions from SOC. It does not relate to any other part of the methodology, except for calculating soil carbon stocks in the encroached area. See the master biomass summary spreadsheets in Appendix B for this conservative calculation.

2.1.3 Project Description Deviations (Rules 3.5.7 – 3.5.10)

1) Data and Parameters Monitored

The parameters $t^{[i-1]}$, $t^{[m]}$, $t^{[m-1]}$, $U_B^{[m]}$, $U_{EM}^{[M]}$ were mistakenly excluded from section 3.3.2 of the validated Project Description (PD) (See Annex B) but are monitored during the crediting period and therefore this omission does not impact the applicability of the methodology, additionally or appropriateness of the baseline scenario. These parameters are included in this MR and will be included in the PD when it is updated at the 10 years' baseline re-evaluation.

2) Biodiversity Monitoring Plan

The Biodiversity Monitoring Plan in section 5.3.1 of the validated PD was described as follows:

Monitoring activities are outlined in the Community Forest Management Agreements, species monitored are IUCN listed species particularly Lion, African Wild Dog and Elephant. Monitoring takes place across the entire project area. Monitoring is done aerially or on foot.

Due to the lack of details of the biodiversity monitoring activities and delayed authorization to conduct these activities within the Game Management Areas from the Department of National Parks and Wildlife (see Approval Letter in Appendix D), the current section's contents are aimed at replacing section 5.3.1 of the PD. The deviation seeks to improve the biodiversity monitoring plan provided, and does not impact on the VM0009 methodology's applicability, additionality and appropriateness of the baseline scenario.

I. Context

Enhanced biodiversity conservation is an essential objective for BCP. The biodiversity monitoring plan must be able to assess the effectiveness of the conservation measures implemented. The monitoring techniques to be used are required to capture adequately trends on a set of indicators by producing robust publishable data while being ground-based, cost-effective and easily repeatable. Additionally, the techniques must be suitable over extensive areas with dense woodlands, rugged terrain and poor road networks. To facilitate the development and implementation of such biodiversity monitoring plans, BCP partnered with the conservation organisation Lion Landscapes¹⁵ (see MOU in Appendix D) and contracted the wildlife and environmental consulting company Western EcoSystems Technology Inc¹⁶.

The resulting plan proposes a layered approach that combines grid-based line transect distance sampling¹⁷ with patch occupancy theory¹⁸ (see BCP Biodiversity Monitoring Plan document in Appendix D). Trends revealed by these methods are further supported by sightings recorded by scouts on anti-poaching patrols using the Spatial Monitoring And Reporting Tool (SMART)¹⁹.

A challenging objective of the current biodiversity monitoring is to provide abundance estimates and trends of species that occur in low densities, occupy large territories and are often nocturnal or elusive such as large carnivores. Actual sightings may be too few to allow estimation of population density by the line-transect distance sampling methods. In this case, basic monitoring of populations is based on estimation of the proportion of "patches" occupied and occupancy modelling (to estimate the probability that a patch is occupied) where a patch is a well-defined grid cell. Trends in abundance can be inferred from this method.

It is important to consider that the outputs of BCP management practices are related to reducing threats to forests and wildlife (i.e. reducing poaching and habitat loss), therefore enhancing biodiversity. However, there are additional factors beyond management control that influence biodiversity conservation, such as drought and disease outbreaks. These factors may result in short term negative impacts on wildlife densities or distribution trends. It is, therefore, essential to consider the long-term trends while analysing annual data. Likewise, it is crucial to monitor trends and distribution of threats (e.g. poaching and habitat loss). Data showing a decrease in poaching and habitat loss will be important in demonstrating management effectiveness, particularly in years when external factors have negative impacts on wildlife densities or

¹⁵ <https://www.lionlandscapes.org/>

¹⁶ <https://west-inc.com/>

¹⁷ Buckland, S. T., D. R. Anderson, K. P. Burnham, and J. L. Laake. 1993. Distance sampling: estimating abundance of biological populations. 1999, Reprint. RUWPA, University of St. Andrews, Scotland.

¹⁸ MacKenzie, D. I., J. D. Nichols, J. A. Royle, K. H. Pollock, L. L. Bailey, and J. E. Hines. 2006. Occupancy estimation and modelling: inferring patterns and dynamics of species occurrence. Elsevier, Amsterdam, Netherlands. Manly, B. F. J. 1997. Randomization, bootstrap and Monte Carlo methods in biology, second edition.

¹⁹ <https://smartconservationtools.org>

distribution. Threats such as poaching and charcoal making are monitored by recording signs of these practices during patrols and line transect surveys.

II. Reconnaissance walks during anti-poaching patrols

Live sightings of animals are collected on a daily basis by scouts across the entire project area, through the Spatial Monitoring and Reporting Tool (SMART). SMART enables the collection, storage, communication, and evaluation of patrol data. BCP has created a data model fitted for collecting, storing and reporting the most relevant data on biodiversity and threats. The compilation of ecological and patrol data from the project area enables a holistic management strategy. BCP also receives technical support from Panthera, one of the nine developers of the SMART tool.

Although the collection of data does not follow a systematic design and no abundance estimates can be inferred from the data collected, the observations are still valuable for monitoring the biodiversity elements as species locations are identified. These observations can be used to direct future systematic surveys within the REDD+ zones. It is also useful in describing demographic variables of key species, such as age, structure and sex ratio when such distinctions can be made.

III. Systematic multi-species survey

a. Survey Areas

Prior to issuing the research permit in February of 2022, that enabled BCP to conduct biodiversity monitoring activities in Game Management Areas, surveys were implemented on private land within the Munyamadzi Game Reserve as the only sampling frame of LCFP. In 2022, two sampling frames were added, namely Shikabeta and Mwanya REDD+ zones (table 4). Ultimately, the objective is to have a set number of permanent and rotating sampling frames that will be monitored over the project lifetime, whose distribution within the project area is representative of land tenure (open areas vs Game Management Areas), management type (Community vs Private), biomes (valley floor vs escarpment/plateau), and wildlife abundance (Low, medium, high).

b. Sampling Frame

A grid of 3km x 3km cells was created for each survey area (see Biodiversity Monitoring Plan in Appendix D for the cell selection criteria; Figure 1). In cases where the sample grid cells to be surveyed is less than 100%, an annual rotating panel design, made of 4 panels is utilised. Panel 1 consists of a number of grid cells that are sampled every year (core panel). Panels 2–4 each consist of a similar number of grid cells that are surveyed on a rotating schedule such that all cells are visited within a 3-year period.

c. Line Transect Distance Sampling

Each grid cell selected receives equal, systematic survey efforts consisting of 2 parallel, north-to-south or west-to-east, 3km transect lines positioned 1.5km apart and 750m from the edges of the cell (Figure 1). By sampling two independent transects within the same cell ensures that the line-transect sampling is aligned with the overarching patch occupancy framework (ie. ≥ 2 independent visits to each sample unit²⁰).

²⁰ MacKenzie, D. I., J. D. Nichols, J. A. Royle, K. H. Pollock, L. L. Bailey, and J. E. Hines. 2006. Occupancy estimation and modelling: inferring patterns and dynamics of species occurrence. Elsevier,

The transect survey occurs annually within each sampling frame. In Munyamadzi, two surveys are conducted, one in early dry season (mid-May to early July) and the other in late dry season (October-November). However, due to logistical constraints, one survey is planned per year for the other sampling frames.

d. Camera Trap Arrays

One or two cameras are placed in the central 1km² of a 3 km x 3 km grid cell. When two cameras are being used, they are placed at the same location, facing more or less each other, on either side of a waterhole, trail, or bait. Two cameras make it possible to capture whisker spot patterns on lions or spot patterns on leopards and wild dogs. This enables the densities of these large carnivore species to be estimated from Capture-Mark-Recapture analysis as well as occupancy modelling from presence/absence data.

The cameras deployed in Munyamadzi are positioned in the field for 12 weeks during the dry season, ranging from early June to no later than October. In 2022, a new camera trap array was deployed in September in the Mwanya REDD+ zone. The monitoring of this new sampling frame is collaboratively managed with the Department of National Parks and Wildlife, using equipment purchased by BCP and donated to the Department (see Deed of Donation in Appendix D).

Table 4. Details of the sampling frames for biodiversity monitoring

Name	Land Tenure	Size (km ²)	No. transects (total length in km)	No. camera trap station	Year of when data collected Started		
					Transect data	Camera Trap data	SMART Patrol data
Munyamadzi	Private	180	38 (114)	24	2016	2020	2020
Shikabeta	Community/ GMA	459	102 (306)	-	2022	-	2020
Mwanya	Community/ GMA	92	46 (92)	20	-	2022	2020

Amsterdam, Netherlands. Manly, B. F. J. 1997. Randomization, bootstrap and Monte Carlo methods in biology, second edition.

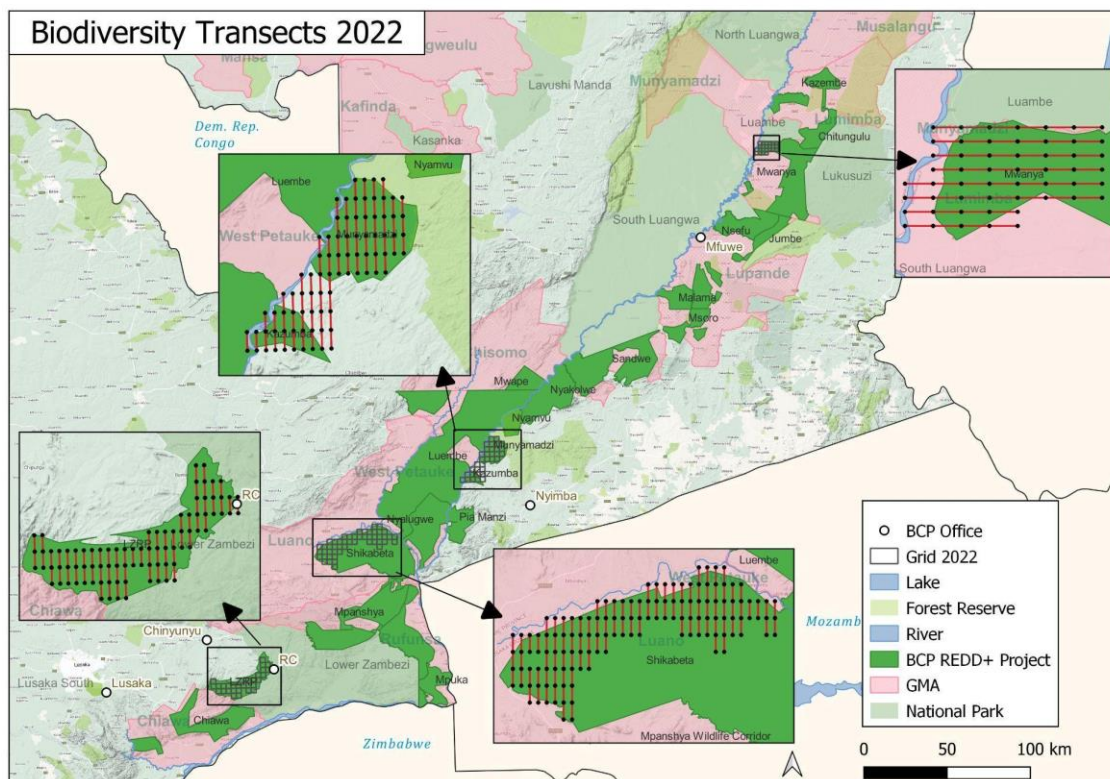


Figure 1. Distribution of the sampling frames for biodiversity monitoring within the BCP REDD+ zones. The red lines indicate transect lines. In each of these sampling frames but in Shikabeta, a camera trap survey is also taking place.

IV. Single-species monitoring

a. Vultures

There are four threatened species of vultures that are found in the project area. Three of these species are listed on the IUCN Red List of Threatened species as Critically Endangered, namely the Hooded vulture (*Necrosyrtes monachus*), the White-headed vulture (*Trigonoceps occipitalis*), and the White-backed vulture (*Gyps africanus*). The fourth species, the Lappet-faced vulture (*Torgos tracheliotos*) is listed as Endangered. Over the last three decades the population of vultures has collapsed across Africa, mainly due to accidental and intentional poisoning²¹. Vultures are an important element of the ecosystem balance as they play a vital role in reducing the spread of disease through scavenging on dead animals that are often infected with deadly bacteria (anthrax and botulism) and viruses (rabies).

BirdWatch Zambia, in partnership with Endangered Wildlife Trust and the programme Vulture for Africa, is contributing to filling the knowledge gaps about vultures in Zambia. This is being achieved through research and monitoring in areas where little is known about the population status and trends of the vulture species.

²¹ Botha, A. J., Andevski, J., Bowden, C. G. R., Gudka, M., Safford, R. J., Tavares, J. and Williams, N. P. (2017). Multi-species Action Plan to Conserve African-Eurasian Vultures. CMS Raptors MOU Technical Publication No. 5. CMS Technical Series No. 35. Coordinating Unit of the CMS Raptors MOU, Abu Dhabi, United Arab Emirates.

In early 2022, aware of the role that LCFP could play, BCP contacted BirdWatch Zambia in order to contribute to its mission through the funding of five tracking devices (see the MoU in Appendix D). In August 2022, solar-powered GPS trackers were cuffed to three White-backed vultures within Munyamadzi Game Reserve. The GPS tracker logs are monitored using the Savannah Tracking tool²² compatible with the EarthRanger platform²³, which provides near real-time movement of the vultures as well as additional vultures from several other funding sources. The movements data are also archived on the MoveBank platform²⁴, which is an online database of animal tracking data hosted by the Max Planck Institute of Animal Behavior.

b. Southern Ground-hornbill (SGH) social survey

The Southern Ground-hornbill (*Bucorvus leadbeateri*) is a large, terrestrial, cavity nesting, carnivorous bird, found across most regions of Southern Africa. However, it is currently listed on the IUCN Red List of Threatened species as Vulnerable, mainly due to extensive habitat loss. Southern Ground-hornbills are co-operative breeders, generally occurring in groups of between 2 and 11 birds. They are a species with a complex breeding biology, with low recruitment rates caused by late maturity, small clutch sizes, and long post fledging dependency. Therefore, there will be slow rates of population recovery where populations have previously been depleted. Due to the vulnerability of the species, and the clear connection between thriving populations of cavity nesting species and healthy forest with large trees, the Southern Ground-hornbills are a key 'trigger species' for BCP's first REDD+ project, the Lower Zambezi REDD+ Project (LZRP). BCP partnered with the Mabula Ground-Hornbill Project (MGHP)²⁵ in order to obtain technical guidance on the monitoring and conservation of this emblematic bird (see MoU in Appendix D).

In addition to the recordings SGH captures during patrols using SMART, a People and Perception Social Survey is conducted in the Chiefdoms of the LCFP. This consists of a questionnaire developed by MGHP to community members that aims to reveal how well the communities know the SGH, and the social and cultural values the SGH may have for them.

V. High Conservation Value Monitoring (HCV)

The HCV assessment followed the guides developed by ProForest²⁶ and the HCV Network²⁷. The three HCV that concern biodiversity attributes include:

HCV 1: Species Diversity. Covers concentrations of biological diversity including endemic, rare, threatened or endangered (RTE) species that are of global, regional or national level significance. Species diversity also covers significant concentrations of biodiversity that are unique or outstanding in comparison with other areas within the same biogeographical unit, at global, regional or national scale.

²² <http://www.savannahtracking.com/>

²³ <https://www.earthranger.com/>

²⁴ <https://www.movebank.org/cms/movebank-main>

²⁵ <http://ground-hornbill.org.za/>

²⁶ Jennings, S., Nussbaum, R., Judd, N., Evans, T., Iacobelli, T., Jarvie, J., Lindhe, A., Synnott, T., Vallejos, C., Yaroshenko, A. and Chunquan, Z., 2003. The high conservation value forest toolkit. Part 3 - Identifying and managing High Conservation Values Forests: a guide for forest managers. Edition I, ProForest, Oxford OX, 12, pp.1-62.

²⁷ Brown, E., N. Dudley, A. Lindhe, D.R. Muhtaman, C. Stewart, and T. Synnott (eds.). 2013 (October). Common guidance for the identification of High Conservation Values. HCV Resource Network.

HCV 2: Landscape-level ecosystems, ecosystem mosaics and Intact Forest Landscapes. HCV 2 covers large significant landscape level forests containing viable populations of naturally occurring species in natural patterns of distribution and abundance at global, regional or national level. It includes forests where ecological processes and ecosystem functioning are completely or relatively unaffected by recent anthropogenic disturbances.

HCV 3: Rare, Threatened or Endangered Ecosystems.

The identification follows a stepwise approach, where the presence of HCV 1-3 qualifying attributes within the project area is investigated through a desk-based review and consultation with local communities, followed by field data measurements when necessary. A comparison is then completed with other areas from the same biogeography to confirm whether or not the HCV 1-3 attributes identified within the project area are relevant to varying spatial scales to have a significant impact on their conservation. The final step is the delineation of the area that requires special management interventions to maintain and enhance the identified HCV, the HCV Management Area.

The HCV attributes identified during desk-based reviews are confirmed through direct patrol sightings and wildlife surveys. This is completed through a participatory survey conducted with community members that have the greatest knowledge of the biodiversity elements of their chiefdoms, namely the community scouts.

3) High Biodiversity Conservation Priority Status

The validated PD (section 5.4.1) lists one species found in the project area that fulfils the criteria of irreplaceability or vulnerability from the Key Biodiversity Area framework, namely the African Wild Dog (*Lycaon pictus*). However, the LCFP qualifies as a site of global significance for biodiversity conservation due to other sympatric species. These species, according to the vulnerability criterion include: White-headed vulture (*Trigonoceps occipitalis*; Critically Endangered, CR), White-backed vulture (*Gyps africanus*; CR), Hooded Vulture (*Necrosyrtes monachus*; CR), Elephant (*Loxodonta africana africana*; Endangered, EN), Lappet-faced vulture (*Torgos tracheliotos*; EN), Grey crowned crane (*Balearica regulorum*; EN), Hippopotamus (*Hippopotamus amphibius*; Vulnerable, VU), Lion (*Panthera leo*; VU), Leopard (*Panthera pardus*; VU), Southern Ground-Hornbill (*Bucorvus leadbeateri*; VU). According to the irreplaceability criterion, the Cookson's wildebeest (*Connochaetes taurinus cooksoni*) endemic to the Luangwa valley, and the Lilian's lovebird (*Agapornis lilianae*), also known as the Nyasa lovebird, are near-threatened species endemic to the Zambezi Basin. The Lilian's Lovebird range is centred within Zambia and more than 5% of the global population located in the Luangwa valley²⁸. Finally, the Thornicroft's Giraffe (*Giraffa camelopardalis* ssp. *thornicrofti*) is listed as Vulnerable on the IUCN Red List of Threatened Species, and endemic to the Luangwa valley, qualifying the project area as a site of global significance for biodiversity conservation.

According to the CCB Standards (Version 3.1, footnote 133, page 48) where several trigger species occur in one project area, it is allowable to focus on at least three species that include (i) species for which populations at the site have the highest global significance, (ii) those facing the most acute threats at the site and (iii) those for which management will be beneficial to a broad range of other trigger species facing similar threats. In the current situation, the species best fitting the selection criteria are (i) the hippopotamus

²⁸ Mzumara, T.I., Martin, R.O., Tripathi, H., Phiri, C. and Amar, A., 2019. Distribution of a habitat specialist: Mopane woodland structure determines occurrence of Near Threatened Lilian's Lovebird *Agapornis lilianae*. Bird Conservation International, 29(3), pp.413-422.

given that the Luangwa River contains the largest population globally; or the endemic Thornicroft's Giraffe and Cookson's wildebeest, and the near-endemic Lilian's lovebird; (ii) vultures given their status as Critically Endangered; (iii) large carnivores (African wild dogs, lions and leopards) given their ecological role as apex predator whose conservation requires healthy prey basis and healthy habitats.

The biodiversity monitoring plan presented above adequately addresses monitoring population status and threats for most of the species. In addition, the conservation measures implemented are beneficial to all species.

2.1.4 Risks to the Project (G1.10)

The major risks, potential impact of risk on climate, community and/or biodiversity benefits and the actions needed and taken to mitigate the risk are described below:

Table 5. List of identified risks to the project with potential impacts and mitigation measures.

Identify Risk	Potential impact of risk on climate, community and/or biodiversity benefits	Actions needed/ taken to mitigate the risk
Political	Government support is important for the longevity of the project.	The project is strategic as a sub-national large-scale project which aims to support and feed into the design of a national REDD+ program for Zambia. Zambia was one of the first nine UN-REDD Pilot Countries in the World with a high level of interest within the government for REDD+ pilot projects. This project is the second REDD+ demonstration project in Zambia and the largest (the first being LZRP which is also implemented by BCP). This project meets the requirements to integrate into the Zambian Governments jurisdictional efforts in the future. The 'GRZ Conservation Account', which BCP implements in LCFP is a project level dedicated account specifically for the Government to access to carry out activities in support of the project. The GRZ Conservation Account currently accrues 5% of net revenue (3% of gross revenue) to fund participation in the project (separate to taxes and permits BCP pays directly to Treasury).

Income Risk	Risk that sufficient carbon revenue is not generated, reducing support of villages	The project's financial viability and breakeven analysis is reported in detail in the Non-Permanence Risk report. The project is set to break even in 2023. The demand for quality offsets has been trending upwards and climate finance is more readily available. BCP is confident of sourcing the most aligned and affordable cost of capital and sale of offsets over LCFP's project lifetime.
Community Risk	Risk that village engagement diminishes resulting in illegal harvest and land conversion to agriculture	BCP's extensive community engagement efforts, strong local presence and partnership with Government have served as powerful tools against these potential project risks. As a result of close involvement of the local community in the design and implementation of activities, community members report that they feel greater ownership of the projects launched in partnership with them. BCP has facilitated many rounds of institutional capacity building training programs for the members of Community Forest Management Groups (CFMGs) and Village Association Groups (VAGs) as local community implementing partner institutions in its REDD+ projects (LZRP and LCFP). These have been followed by an ongoing monitoring program. BCP is improving the CFMG capacity to manage funds, design and implement annual work plans. Under the field-based extension model, BCP has a consistent presence in local communities with regular site visits and monthly monitoring. "Community Forest Management Partnership Agreement" (CFMPA) between CFMGs and BCP are long-term agreements that provide a standard in which performance can be monitored, measured and tracked. Zambia benefits from one of the most advanced legislative enabling frameworks for community forestry in the world, in which communities can apply for forest user rights, including carbon. BCP and its Government partners have supported the first CFMGs to apply for the form 7 step as a step towards obtaining forest user rights, including carbon. This is fundamental as research clearly indicates that communities are incentivized to manage what they own.

	Substantial and repeated crop failure in surrounding communities could potentially lead to increased poaching and charcoal production for financial benefit. More erratic rainfall and pests make this likely.	The alternative economic development efforts established by BCP are aimed at mitigating this risk, by promoting more resilient conservation agriculture techniques, and supporting alternative, sustainable livelihoods activities.
	Human Wildlife Interaction (HWI) may cause conflict and therefore create social risk. As the project succeeds at restoring wildlife, there is a risk that conflictual HWI could increase, leading to livelihood damage, risks to safety, retaliatory killings, and resentment towards conservation efforts.	BCP is building a set of best practice tools to mitigate HWI conflicts, including strategic fencing capabilities, livestock protection, and crop protection measures.
Fire and drought	Fires can cause extensive damage to property, crops, forests (areas of carbon stocks) and loss of lives to both humans and animals. The threat of fire to the project's carbon stocks, the Miombo and Mopane woodlands, and their associated biodiversity are discussed in detail in the Non-Permanence Risk report (Appendix L).	BCP implements fire management, primarily in the form of firebreaks and early burns. These mitigation strategies are conducted in partnership with neighbouring communities, with an aim to prevent potentially damaging, hot, late season fires. An early burning fire management program mitigates the risks that droughts pose in contributing to late season fires that are potentially damaging to vegetation.
Elephant damage	Elephants could impact woody species density negatively.	As the current elephant population does not pose a threat to the project in terms of damage to the climate, community and biodiversity benefits, no specific mitigation measures were implemented during this monitoring period.

Natural Disasters	Examples include agricultural events such as the armyworm pest that was declared a national emergency in 2012 and again in 2016; floods, droughts, etc. Such types of natural disasters could result in illegal harvest and land conversion to agriculture.	With a view to mitigate the risk of natural disasters, the BCP's community projects are designed to promote and build local stakeholder resilience to disasters. This includes the Projects such as crop diversification, income diversification, and conservation farming training program, which is designed to ensure that improved farming techniques are implemented. LCFP also focuses on clean water provision therefore reducing the risks of scarcity, or contamination, in the possible events of floods or droughts.
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2.1.5 Benefit Permanence (G1.11)

The project has been carefully planned to become self-sustaining and an engine of green economic growth. A key strategy to ensure that the project's benefits last beyond the project lifetime is capacity building for local communities. BCP has facilitated many rounds of institutional capacity building training programs for the members of CFMGs and VAGs as local community implementing partner institutions in its REDD+ projects (LZRP and LCFP). Additionally, BCP is improving the CFMG capacity to manage funds, design and implement annual work plans. Promoting local involvement in and ownership of projects, as well as to building local capacity is expected to promote a self-sustaining project. The project could then take advantage of opportunities beyond the 30-year project lifetime, such as new revenue opportunities from the evolution of water and biodiversity credits as Payment for Ecosystem Services (PES) projects.

The project also aims to develop wildlife breeding and restocking capacity to restore areas for tourism and live game sales, through initiatives implemented in partnership with the Department of National Parks and Wildlife (DNPW) and private game ranch.

Finally, by supporting CFMGs to apply for the form 7, BCP contributes to the long-term sustainable use of the forests as research clearly indicates that communities are incentivized to manage what they own. During the monitoring period, 17 CFMG's have applied for form 7 and obtained forest user rights.

All these measures will ensure that the project's benefits will last beyond the project lifetime.

2.1.6 Grouped Projects

For grouped projects, Environmental and Social (E&S) risk management has been integrated into BCP's operations on two levels, company and REDD+ project. On the project level, E&S risk management will be integrated into the BCP REDD+ project development cycle, in order to ensure that E&S considerations are fully considered during the project design, development and implementation.

1) New Project Areas and Communities (G1.13)

The second project activity instances comprise a total area of 92,990 ha, of which 55,351 ha consist of forest. Within this forested area, 50,730 ha is calculated as project accounting area (Table 6 and 7) and subject to verification in this monitoring period.

Table 6. Summary of Project Area REDD+ Project Zones on Private Land, Second activity instances.

Private Game Ranch	Chiefdom	Project area (ha)	Forest area (ha)	Accounting area (ha)
Nyakolwe	Luembe	57,706	32,013	29,856
Nyamvu	Luembe	15,238	7,176	7,172
Kazumba	Mwape	4,775	2,732	2,724

Table 7. Summary of Project Area - REDD+ Project Zones under communal tenure, Second activity instances.

Community Forest Management Group	Game Management Area	Chiefdom	Project area (ha)	Forest area (ha)	Accounting area (ha)
Mpanshya corridor	Rufunsa	Mpanshya	15,271	13,430	10,978

The third project activity instances comprise a total area of 199,351 ha, of which 170,636 ha consist of forest. Within this forested area, 92,313 ha is calculated as project accounting area (Table 8) and subject to verification in this monitoring period.

Table 8. Summary of Project Area - REDD+ Project Zones under communal tenure, Third activity instances.

Community Forest Management Group	Game Management Area	Chiefdom	Project area (ha)	Forest area (ha)	Accounting area (ha)
Kazembe	Upper Lumimba	Kazembe	18,515	11,246	8,851
Mwape	Chisomo	Mwape	20,500	16,914	4,247
Chiawa	Rufunsa	Chiawa	66,250	61,637	30,682
Mphuka	Rufunsa	Mphuka	25,881	22,584	17,586
Mburuma	Rufunsa	Mburuma	68,205	58,255	30,947

2) Project Zone Map (G1.13)

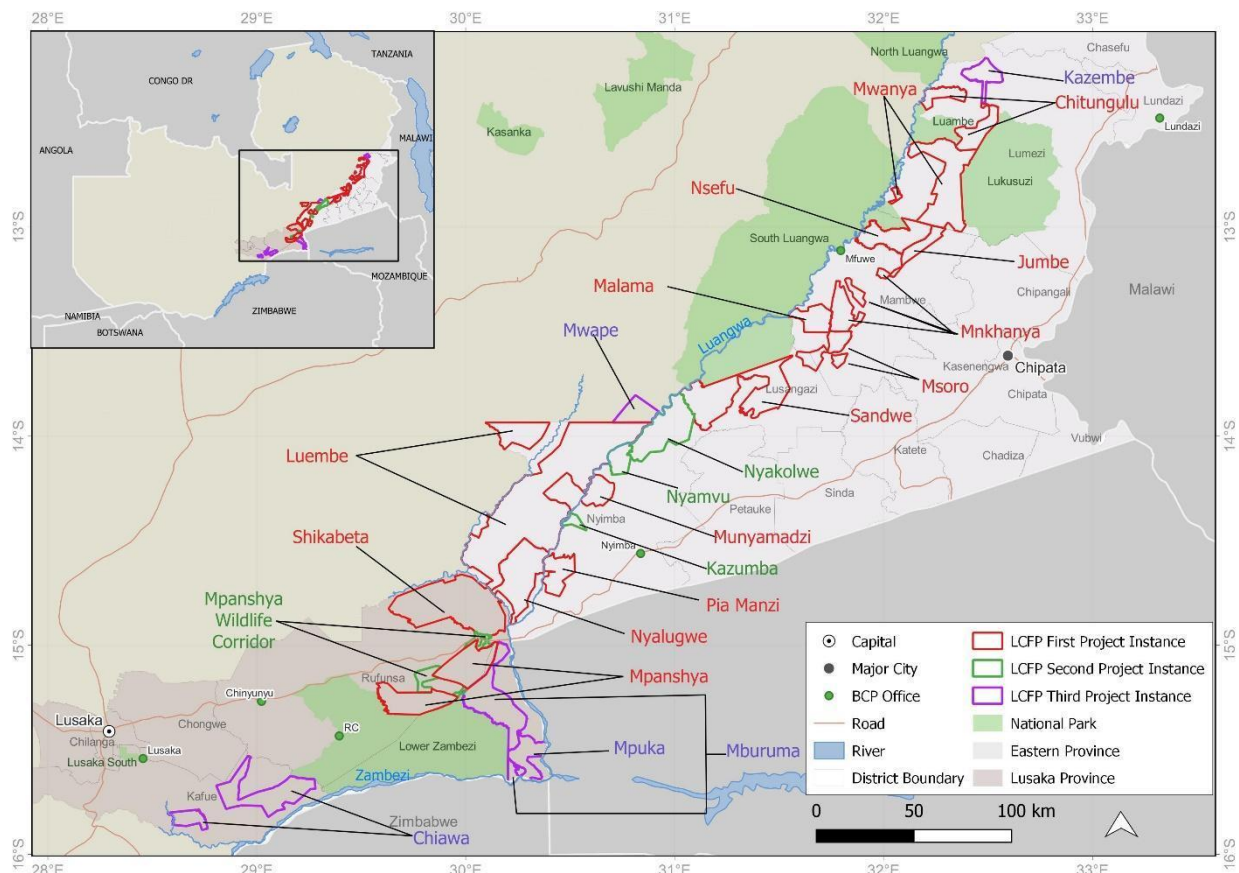


Figure 2. LCFP first, second and third Project Activity Instances

3) Eligibility Criteria for Grouped Projects (G1.14)

Following the guidance of grouped projects under the VCS, all the new project activity instances shall meet the eligibility criteria for inclusion as explained in the PD, section 2.1.20 as shown below in Table 9.

Table 9. Eligibility criteria with the second and third project instance incorporated into the project.

Criteria	Justification	Proof (Relevant Documents)
Must be within the chiefdoms already identified as potential future activity instances in the PD.	All the project instances are within the project boundaries identified in the PD (page 36-40).	Map showing project boundaries (see Figure 9 & 10 and Tables 4-6 in the PD).
Future project instances must be avoided unplanned deforestation. Accordingly, the end land use in the baseline scenario must be non-forest.	The baseline scenario and the demonstration are determined for the entirety of the geographic project area within which project activity instances are developed.	PD, Figure 10, page 40.

Criteria	Justification	Proof (Relevant Documents)
The baseline drivers and agents of deforestation in future project instances must be the same as current project instances - clearing of forested land for expansion of small-holder farming and, or charcoal production.	All areas share the same agents and drivers as the first project instance and therefore share the same conclusion.	Drivers and agents study by UNIQUE (Appendix B).
Future project accounting areas must have met the Zambian definition of forest for at least 10 years prior to the project start date.	All new instances meet the definition of forest and accounting areas were calculated according to the same method as First Activity Instance areas.	Classification method described in the PD.
Within 2km of the future accounting areas perimeter, at some point a forest must have been converted to farming land.	Perimeters of the newly added areas under the second and third project activity instances satisfied the requirements without exception.	rU value in NER calculations in Appendix B.
Future project accounting areas must be located in a tropical ecosystem and must not contain peat soil.	No peatland was detected in the project areas.	Spatial data for accounting areas does not overlap with available peatland data (See Appendix C).
For each new accounting area, a reference region meeting the requirements in terms of size and similarity to the accounting area as per section 6.8.1 of VM0009 v3.0 must be delineated.	First instance project areas share a common reference area as the new instance.	Map and spatial data for the common reference areas.
For any new reference regions that may be required, there is sufficient historical imagery meeting the requirements of section 6.8.4 of VM0009 v3.0.	Second and third instance project areas share a common reference area as the first project activity instances, and therefore quality imagery is available.	Map and spatial data for the common reference areas.
Project activities for new project instances are the same as those described in the PD.	NA	NA

Criteria	Justification	Proof (Relevant Documents)
The project can access any new leakage or proxy areas that may be required for monitoring related to new project instances as per sections 8.3.2.1 and 6.4 of VM0009 v3.0.	These areas share common leakage and proxy areas with the first project instance areas.	Map showing accessibility to the common leakage and proxy areas – see MR5 Figure 9.
Same Governance, Land Tenure and Social Economic Systems.	All second and third instance activity areas are private and communal land tenure systems.	Title Deeds from private game ranch owners and community forest management Agreements signed between government and CFMG's.

A section of 19,887 ha of the new second instance project activity areas overlapped with the existing activity, thereby shifting the leakage area (originally 319,179 ha), to 299,291 ha. However, this is not regarded as a methodological deviation, as VM0009 allows for portions of existing leakage areas to be converted to future project activity instance areas.

4) Scalability Limits for Grouped Projects (G1.15)

Scalability limits are primarily land use and administrative boundaries. In relation to LCFP these include the Mozambique and Malawi borders, and national park boundaries, whereby new project activities are not permitted to take place beyond these limits. All new project instances added since project validation are within these limits. In addition, there is no anticipated limit beyond which the activities implemented within the frame of the LCFP would stop generating net positive climate, community or biodiversity benefits.

5) Risk Mitigation for Grouped Projects (G1.15)

The addition of new project areas is not anticipated to result in risks to the project's net positive climate, community or biodiversity benefits. Furthermore, BCP developed an Environmental and Social Management System (ESMS) that is implemented at the company and project levels. All risk mitigation and environmental and social (E&S) procedures follow the guidelines set out in BCP's ESMS Manual (see Appendix L) and are initiated during the viability stage of project development. As BCP adds new sites to existing projects, the E&S procedures start at site-level development (Step 3, pre-agreement), and include any required E&S assessments. In order to mitigate risks arising from grouped projects, BCP has designed an E&S risk management process. This process aims to integrate the identification, assessment, and management of risks and impacts. This includes monitoring and evaluation of environmental and social performance, into the REDD+ project development cycle. The E&S steps have been aligned with the project development cycle, as illustrated below.

Step 1 – Project landscape E&S screening, during viability phase;

Step 2 – Project landscape E&S assessment and provisional E&S management plans, during feasibility phase;

Step 3 – Site-level E&S assessment and E&S management plans. Final project level E&S management plans, during development (pre-agreement) phase;

Step 4 – Project and site level E&S management plans implemented, during development (post-agreement) phase;

Step 5 - On-going implementation, M&E, during implementation phase (post-agreement).

6) Changes to Management (G4.1)

The incorporation of new project instances has not affected the current management structure.

2.2 Stakeholder Engagement

2.2.1 Stakeholder Access to Project Documents (G3.1)

The PD and monitoring reports are available in hard copy and distributed to each participating chiefdom's CFMG Office, where community members and government partners have free access. In addition, district Commissioners in respective project locations and offices of the Principal Forest Officer in Eastern and Lusaka Provinces also hold copies of these documents.

Subsequent monitoring reports between 2018 - 2021 were shared through informal meetings with project stakeholders in each of the 17 (including additional instances) participating Chiefdoms. Due to high levels of illiteracy, informational materials are shared. Presentations and meetings are led by local staff and implementing partners, who are trained to accurately describe the content and significance of the PD, verification process, and to respond to any questions or concerns that may arise by community stakeholders.

2.2.2 Dissemination of Summary Project Documents (G3.1)

The project PD and subsequent monitoring reports are shared through informal meetings with project stakeholders in each of the 17 participating Chiefdoms. Due to high levels of illiteracy, the executive summaries of the respective documents have been translated into local languages. Additionally, the dissemination of these informational materials is combined with presentations and meetings led by local field staff and implementing partners. These representatives have been trained to accurately describe the contents of the PD and monitoring reports, explain the verification process and the significance of these documents, and respond to any questions or concerns that community stakeholders may raise about the project.

Translated climate and biodiversity monitoring plans and impact sheets have also been distributed to the chiefdoms.

2.2.3 Informational Meetings with Stakeholders (G3.1)

Informational meetings have been held in each of the 17 partner Chiefdoms of LCFP, with mechanisms in place for further information dispersal and facilitation of community stakeholder feedback at village and VAG levels (See in Appendix J).

Community meetings are publicly announced through various ways. These include posting public notices at central locations (schools and clinics) and closely involving the CFMG to mobilise meetings in advance as well as through traditional leadership methods.

2.2.4 Community Costs, Risks, and Benefits (G3.2)

Potential costs, risks and benefits of the LCFP REDD+ project model have been widely distributed to all local partner chiefdom communities throughout the project during the FPIC process and through community engagement meetings (Appendix E, J and K). The risks discussed include market risks that affect carbon credit prices that are beyond BCP's control. While communities might not necessarily understand all the aspects related to carbon sales, there is awareness that payments can go up or down depending on a global market. Discussions about costs, risks and benefits are an important part of the FPIC stakeholder engagement processes that took place between 2014-2022. BCP partnered with the government at all stages. Meetings held with stakeholders in all 17 partner Chiefdoms were documented, these include a combination of Chiefdom, VAG (based on the 20-km polygon criteria specified in the CCB Standard) and village level meetings. Individual sensitization activities targeted at key stakeholders and focal groups (those living along or near to proposed protected forest area boundaries) were also documented.

Information was presented, and stakeholder feedback was solicited throughout a series of participatory meetings that took place following the PRA/PLA techniques, in each of the partner Chiefdoms, as part of the forest identification process.

The forest selection process was stakeholder-led, involving representatives from community institutions (CFMGs, VAGs), as well as traditional leaders and local Government partners. The process was facilitated by trained BCP staff members (FPIC and PRA/PLA trained) and was well documented. Program process records were made available to demonstrate that community leaders were sufficiently involved, the program was fully described and adequate information was provided to allow community members to make informed decisions about the forests identified for protection through REDD+ activities. This information was reiterated at the time of signing community consent letters for REDD+ implementation (at the end of 2016, following the completion of nearly 2 years of stakeholder sensitization and engagement activities), signing Conservation Fees Guidelines (early/mid 2017), and signing Conservation MOUs (at the end of 2017).

Stakeholders have experienced approximately one year of consistent activities and further information to understand the practises and implications of undertaking REDD+ activities. This has allowed for informed ongoing decisions about consenting to ongoing REDD+ implementation. Community stakeholders have continued to remain supportive of REDD+ activities within their Chiefdoms is a strong testament to their ongoing understanding of, and support for, the project taking place.

2.2.5 Information to Stakeholder on Verification Process (G3.3)

Stakeholders have been informed of verification processes through a series of meetings that have been held at a Chiefdom-level, in each of the 17 partner Chiefdoms. These meetings have been held by trained members of staff, as well as implementing partners, including representatives from local Government and community institutions (such as CFMGs). The meetings have followed a standard curriculum, with the aim of ensuring that key points are accurately and consistently communicated in a standardised way. Information is shared in local languages, in locally appropriate ways. Informational materials (handouts) outlining/explaining the verification process have also been developed and distributed widely through local community institutions (Appendix J). As VCS verification processes by independent V/VB have been taking place on a yearly basis since the validation of the project in 2019, the procedure followed to inform the stakeholders is in place and will also be followed for the CCB Verification.

2.2.6 Site Visit Information and Opportunities to Communicate with Auditor (G3.3)

Similar to the process followed for verification of BCP's first REDD+ project (LZRP) and validation of LCFP, BCP will ensure that the schedule for auditor's site visits, and community meetings, is widely announced - both through verbal announcements disseminated through local staff, implementing partners and community institutions, as well as through hard-copy announcements that will be posted openly in public spaces (Appendix J).

The announcement process will be documented through photographs of announcements made publicly in communities, in advance of meetings and/or visits. As per the CCB program rule, the communities will be informed of the auditors' visit preferably 30 days in advance.

2.2.7 Stakeholder Consultation (G3.4)

Stakeholder feedback is critical, especially since BCP's REDD+ model, under the LCFP, is to empower and support local community and Government partners to implement REDD+ activities in the long term (see Appendix J) BCP has ensured effective consultation in the following ways:

- Stakeholder-led identification of selected areas and boundaries for forest protection, based on a full and accurate understanding of "what it means" to protect an area, under REDD+ (including: potential costs and benefits, as well as allowed / not-allowed activities). BCP is invested in an innovative methodology called 'Flying Fpic". This methodology was specifically designed by BCP and partners for application in LCFP. The challenge identified was that some Community Forests identified for first conservation were vast and had little road access. As a result, walking or ground-truthing the entire boundaries would have been very difficult within the project timeframes and implementation plan.

Flying FPIC involves filming the forest and boundaries selected by communities, and displaying these back in a cinematic style. It is designed to replicate the flight experience as many community members have not flown in a plane prior and to minimise risk of disorientation. Community members and Government officials are then flown systematically around the proposed Community Forest to look for unseen settlements, and are given the opportunity to refine the Community Forest Boundary. The community then selects areas of sensitivity to 'ground truth' which are targeted on site visits to refine and adjust boundaries, and conduct additional onsite FPIC with households living near a proposed Community Forest boundary. This method is effective at locating villages that may have moved into a forest without CFMG knowledge, as well as to demonstrate corridor connections that could be developed. The aim is to ensure that community stakeholders continue to remain supportive and informed of REDD+ activities being undertaken in their Chiefdoms.;

- Participatory identification of critical community development needs. BCP and the CFMG's have collaborated to implement the needs assessment with each of the VAGs in order to identify community projects and livelihood activities for the following years, with direct inputs from community members/VAGs. Women's priorities are well reflected in projects selected as a part of the annual planning process.
- Collaborative development of an appropriate forest monitoring model, to be undertaken by CFMG member's/community scouts, with support from BCP and through the REDD+ project;
- Participatory identification of critical capacity building needs, to empower community and government stakeholders to implement REDD+ activities on their own.

Overall, this community-led approach in regards to project design and implementation of REDD+ has contributed to a sense of unity among community members in relation to project activities.

2.2.8 Continued Consultation and Adaptive Management (G3.4)

BCP's approach to continuous consultation and promoting adaptive management include annual collaborative work planning with CFMGs, capacity building of CFMGs, and participatory monitoring and evaluation of community projects and resource protection, as described in details hereunder.

The participatory Annual Work Planning process undertaken by BCP and CFMG's began by evaluating the performance of previous year goals and objectives. BCP and the CFMG's then collaborated to implement the needs assessment with each of the VAGs in order to identify community projects and livelihood activities for the following years, with direct inputs from community members/VAGs. Using a problem tree analysis community projects were selected, budgeted and planned in the CFMG Annual Work Plan.

As part of the CFMG forest monitoring plan and resource protection mandate, BCP and the CFMG's collectively identified the key areas and activities for continued resource protection, including but not limited to:

- Ongoing support to community scouts;
- Improved fire management/early burning and limiting late dry season fires;
- Reviewing forest boundaries and ensure they are communicated to community members;
- Proactively address encroachment and potential deforestation of community forests;
- Sensitise and build awareness on Human Wildlife Interaction (as opposed to human wildlife conflict)

BCP's capacity building of CFMG's is focused on conducting a participatory needs assessment with the CFMG's and subsequently facilitating inclusive workshops and meetings to build on common understanding and expectations between CFMG's and BCP around 5 key topics:

- Project and financial management;
- Roles and responsibilities of CFMG's;
- CFMG formation and governance;
- REDD+ training and awareness building;
- Grievance procedure.

Finally, BCP is committed to conduct ongoing participatory monitoring and evaluation of projects in collaboration with the CFMG. This process includes regular project visits to track the progress of community projects and to evaluate the challenges in implementing the projects with CFMG members. These collective site visits and progress reports are critical for enhancing project management and timely achievement of project completion. Ongoing communication and consultation are taking place between BCP and local community and Government stakeholders, through the following mechanisms and channels:

- Locally hired BCP staff from each partner Chiefdom - provide 24/7 communication channel;
- Regular coordination visits between BCP team leaders and community institutions and implementing partners under BCP's operational model. Due to regular site visits, opportunities for meetings with traditional leaders and/or members of the wider community arise if they are requested by either side;

- Quarterly partner meetings taking place at a District level, and potentially Chiefdom-level, involving representatives from each Chiefdom/community institution, traditional leaders, local Government partners, and any other local implementing partner;

All of the above are “internal” channels for communication and collaboration among REDD+ implementing partners for the LCFP. In regards to communication with a wider range of external actors, the following mechanisms are in place:

- Members of the wider community may participate in ongoing sensitization meetings, and/or quarterly partner meetings, which are open to the public;
- BCP posts regular update blogs to the website – open to the public;
- BCP uses social media (Facebook, Twitter, Instagram) to disseminate key messages and updates – which are open to the public;
- Information is regularly updated and available on the BCP websites²⁹ (<https://www.bcpimpactmonitoring.com/>; <https://bcp.earth/>);
- Informational materials such as handouts and flyers about REDD+, BCP and CFP are available in hard and soft copy, upon request.

2.2.9 Stakeholder Consultation Channels (G3.5)

To ensure legitimacy of the consulting process and have legal and legitimate consultation channels, communities are required to form CFMG to be eligible to enter a Community Forest Management Agreement (CFMA) and gain access to Community Forest Management as provided under the Forest Act of 2015.

The functions of the community organisations include:

- Providing a legal entity that can enter into agreements with other partners;
- Providing a point of entry to provision of management, capacity building and funds;
- Providing a forum for the resolution of disputes within the community;
- Ensuring equitable sharing of resources within the community following the signing of a CFMA; and
- Providing community members with capacity building in organisational, financial and enterprise management.

MOU's have provided channels of communication and terms of engagement, defining roles of each party in the LCFP. From the start of the project, community consultation was documented and channels of communication were established.

FPIC was used as a tool to enhance information sharing and provide as much information as possible to communities.

The FPIC information sharing guidelines are:

- Approach government at National, Provincial and District level

²⁹ <https://bcp.earth/>; <https://www.bcpimpactmonitoring.com>

- Community request for viability assessment
- Consent to REDD+ implementation request from Chieftdom after completion of the viability assessment
- Participatory forest selection using Participatory Learning Approach (PLA) tools conducted
- Start date letters written, preparing communities to enter the CFMA

2.2.10 Stakeholder Participation in Decision-Making and Implementation (G3.6)

BCP's FPIC Guidelines have been designed to ensure participation of all stakeholders on an equal basis (Appendix J). The consultation procedure ensures that all members of all communities have an opportunity to be present at meetings and that benefits are equally distributed. Records are kept of all meeting participants by gender and made available to the VVB.

In addition to following the FPIC guidelines to regular consultation meetings, BCP also ensures participation through the collaborative annual work planning process described in 2.2.8. In addition, existing community governance structures are utilised for project implementation. Forest monitoring is undertaken by CFMG member's/community scouts, with support from BCP. BCP supports the employment of female Community Scouts who have passed shortlisting, selection, basic, advanced, and in service training, as well as male Community Scouts. As a result, around 30% of BCP-supported Community Scouts are women (as of November 2022, 23 out of 78 scouts are women).

2.2.11 Anti-Discrimination Assurance (G3.7)

BCP has a Ministry of Labour approved HR Policy that follows Zambian labour laws and international standards. BCP policies do not allow any form of discrimination or abuse by any employees or project participants while participating in project activities. BCP has a policy entitled "Workplace Harassment, Discrimination and Bullying Policy", and extends to contractors.

Other entities involved in project design, such as CFMGs, are Government recognized, and therefore are required to comply with Government standards to mitigate discrimination and sexual harassment. As mentioned above, BCP supports the employment of female Community Scouts. Community Scouts are all held to strict DNPW and CFMG codes of conduct as they are required to work under the supervision of Government Wildlife Police Officers, who are also governed by strict uniformed services policies against discrimination and sexual harassment.

Therefore, organizations from the Government involved in this project (DNPW and Forestry Department) are governed by set Government policies within the laws of Zambia to ensure that civil servants are not complicit in discrimination and sexual harassment. This also applies to CFMGs which fall under the Wildlife Act of 2015.

In order to mitigate against possible discrimination to project staff and partners, BCP's agreements with communities consistently declare that the definition of non-performance is; 'threats, violence towards or blocking of access of BCP staff and partners to areas of work'.

2.2.12 Grievances (G3.8)

BCP treats all grievances regardless of their origin with urgency. BCP anticipated that whilst the implementation of the LCFP would have positive impacts on the lives of the rural communities and the

environment, there would nonetheless be some grievances from the communities. To address these, the LCFP implemented a grievance mechanism that was communicated with communities and implementing partners (refer to Appendix L). To align to international best practices, social safeguards and Zambian legislation, the LCFP in 2022 revised this grievance mechanism to ensure that communities are provided with a means to address their grievances without the fear of being reprimanded (refer to the revised stakeholder grievance mechanism in Appendix L). The Stakeholder Grievance Mechanism (SGM) of the LCFP operates under a decentralised approach to ensure that grievances are captured from the lowest possible levels of project activity implementation. A copy of the grievance policy is available at the CFMG offices.

The SGM has been revised to include seven guiding principles; equity, accountability, transparency, accessibility, anonymity, timely response and confidentiality. In developing this revised stakeholder grievance mechanism, the LCFP has taken into consideration and incorporated the existing informal and formal structures of dispute resolution to enhance effectiveness and inclusion whilst respecting the local cultures and customs of the respective areas thereby ensuring that the mechanism is easily accepted by the majority of stakeholders.

In the revised stakeholder Grievance Mechanism, the LCFP apart from receiving written grievances also encourages community stakeholders to raise any questions or concerns that they might have about the project with community-based project staff, including: Chiefdom Assistants, Chiefdom Officers, Agriculture Officers, Chiefdom Leads, Sector Managers, Conservation Managers, Conservation Officers, Project Coordinators or other BCP staff. In the event that the LCFP field staff are not able to provide immediate feedback, the project field staff will capture using forms provided in the revised grievance redress mechanism and log in the grievance onto the LCFP grievance register for tracking investigations until the grievance is closed out and communicated to the community in writing and addressed during community meetings. The Project guarantees that it will respond to all grievances within 30 days of receipt.

There were 68 grievances received from 2019 to 2022 across the 17 Chiefdoms. The details of each individual grievance received since 2019 is available in the grievance registers in Appendix B and Appendix L. All grievances are closed with the exception 4 grievances; 1 from 2021 and 3 from 2022 which related to:

- a. the 2021 grievance received from Nthumbe VAG in Chitungulu Chiefdom on the distance of community fields from the REDD+ Zone; In the process of trying to resolve this grievance, the Traditional Leader; HRH Chief Chitungulu passed on and therefore, this grievance was placed on hold until a new Chief is installed. This was communicated by Project staff to the community and the grievance will be resolved after a new Chief is installed to oversee management of customary land and consent to any adjustments to Community Forest areas (REDD+ Zones).
- b. 2022 grievances received from;
 - Mwanya Chiefdom; on the mis-appropriation of community forest carbon fees by the CFMG; This grievance is still being investigated by the Department of National Parks and Wildlife, which has resulted in the suspension of some members of Mwanya CFMG and transfer of authorising officers from the South Luangwa Area Management Unit to pave way for full investigations.
 - Luembe Chiefdom relating to a grievance received from Chikwasha VAG on the distribution of livelihood support to communities by Luembe CFMG in the form of food rations. The Project staff

have instituted investigations and officially communicated the receipt of this grievance to the CFMG Patron; HRH Senior Chief Luembe and Chairperson. This grievance is still being investigated.

- Mpanshya Chiefdom; CFMG Chairperson who felt that project staff conduct during community sensitisation meeting specifically how topics on gender mainstreaming in project implementation was targeted directly against him and bound to affect his claim for re-election. This grievance is still being addressed by the project staff.

2.2.13 Worker Training (G3.9)

BCP has a well-developed training programme for employees and community members (see below), starting with an initial 2-day long on boarding period in BCP's Headquarter. New staff members are provided with Human Resources (HR) training, to ensure that they are aware of their rights and responsibilities as employees. The BCP Standard is introduced to new staff which is a repository of BCP institutional knowledge, key policies, positions and a guide to succeeding at BCP (Appendix A).

Members of BCP's community-based Forest Monitoring Team, for example, have received training from senior managers (including BCP's carbon managers), and have access to handbooks and SOPs that have been specifically created to provide on-going information and training for employees. Similarly, members of Community Engagement Teams receive training on the FPIC process (see BCP's FPIC Manual V8.0 in Appendix J).

The knowledge acquired during the on boarding period is maintained through regular refresher training. Likewise, community members are being educated repeatedly on the relevant topics through the multiple meetings taking place on a monthly basis in all partner chiefdoms. Additionally, REDD+ training has been offered to Government counterparts.

Furthermore, BCP operates an annual "Inspiration Fund" where approximately \$20,000 per annum is designed as a pool for BCP staff to further their professional development. Recent awardees have completed Harvard Business School management courses, Master's degrees and Bachelor's degrees, HCSA-HCV assessment training courses (Appendix I).

Below, is a list of training courses that have been presented to BCP staff either by in-house experts or experts from partner organisations:

- **Compliance with Socio-Economic Standards of the Verified Carbon Standard (VCS) and Climate Community and Biodiversity Alliance (CCBA) Standards** - To ensure field extension staff are familiar with the requirements of the above standards and collection of the correct field data, as well as follow the correct protocols in dealings with local stakeholders.
- **Forest Biomass Data Collection in Compliance with Carbon Market Standards** - To train local community members to collect the required forest biomass data for baseline projection purposes as well as future compliance monitoring
- **Wildlife and Forest Law Enforcement Training** - To facilitate DNPW to train community scouts operating within REDD+ project areas in the enforcement of forest and wildlife laws to ensure adequate protection of core project areas.

- **GIS, Remote Sensing and GPS use for Forest Carbon Projects** - To train key BCP field staff in the use of Geographic Information Systems (GIS), Remote Sensing (RS) basics and Global Position System (GPS) use in order to streamline compliance of projects with carbon market standards.
- **Principles of Deforestation, Climate Change, REDD Mechanisms and BCP's Model** - To train Community Engagement Team (CET) members in the principles of climate change and REDD+ mechanisms model. The CET also works to empower, accurately describe and promote BCP's work at a community level, facilitating sensitization, project support and decision-making. After undergoing this training and gaining relevant community experience, members of BCP's CET present an abbreviated program to local communities, to raise awareness of the significance and implications of a REDD+ project, in order for BCP to obtain free, prior and informed consent (FPIC) from local communities. CET members are explicitly instructed to target and involve members of under-represented groups in their sensitization and consultation efforts.
- **Advanced driver training:** All BCP staff authorized to drive company vehicles have received advanced driver training from a recognized driving institute. The training included a lecture session which covered defensive driving and risk prevention. It was followed by vehicle inspection procedures, road driving tests and off-road driving. Off-road driving included safe vehicle recovery and four-wheel drive systems.

The following training have been provided to community members involved in the project activities:

- **Conservation farming** - All participants in the conservation farming project receive training on improved farming techniques which includes: Promotion of crop residue retention; Promotion of minimum soil tillage; Promotion of crop rotation; Promotion of agroforestry; Promotion of the use of reduced chemical fertilizer and increased organic manure.
- **Bee keeping** - All beekeepers and mentors received original training from Bee Sweet and subsequently from Nature's Nectar before embarking on the endeavor.
- **Biodiversity monitoring** - In-house biodiversity monitoring training has taken place annually since 2020. Members of the Forest Monitoring Team, general workers, Logistics Technician and scouts were trained in animal identification and use of appropriate devices (Android devices for data capture using the Spatial Monitoring and Reporting Tool, SMART) and equipment (GPS handsets, range finder, compass). Before each line transect distance sampling survey, the team receives a 2-day refresher training. In 2021, all scouts received a 2-day training on animal track and sign identification by external experts from Bedrock Africa and Chipembele Wildlife Education Trust in Mfuwe.

2.2.14 Community Employment Opportunities (G3.10)

A fundamental aspect of BCP is to create employment for communities within partnered Chiefdoms. BCP tracks the percentage of staff from partnered Chiefdoms on the payroll on a monthly basis as part of BCP's Monthly Management Indicators. As in November 2022, this figure stood at 45% of approximately 242 people on payroll (including Community Scouts) (see Appendix I). BCP encourages and seeks Chiefdom hires who meet qualifications (encouraging women to apply), and actively builds Chiefdom colleagues through consistent investments in capacity building. All staff employed have training specific to their positions e.g. scout training. When jobs cannot be filled from local communities, recruitment then becomes nationally advertised through Zambian websites, social media and BCP's website. The entire Forest Monitoring Team, Community Scouts, some community engagement staff, majority of the support staff are all local project zone members.

BCP adheres to all employment equity and labour laws and places an emphasis on employing local community members. Job opportunities are widely advertised and publicised amongst local community members. BCP aims to pay above Zambian statutory minimum wage and embarks upon regular benchmarking exercises to ensure entry level staff meet or exceed comparable organisations in the same sector in Zambia. As a result of this approach and commitment, among other things, BCP is B Corp certified ('benefit corporation') since 2017. BCP is an equal opportunity employer and prides itself in giving all people an equal opportunity for employment.

BCP's job application process is straightforward and transparent. When announcing positions, BCP releases a job advertisement that outlines the position, expectations, qualifications required and clearly states that BCP is an equal Opportunity Employer. To apply for a position, candidates submit a CV and cover letter/letter of interest to BCP. In the case of community applicants, a less formal list of qualifications is required, and a BCP staff member is able to assist the applicant in cases of illiteracy. In some cases, BCP will conduct a trial of several days for interested applicants for practical positions such as an engagement position. This is to gauge skills and attitude fit with BCP's team, and enable the candidate to make an informed choice about committing to the field and community nature of BCP's work. BCP clearly disseminates information about the place, date and time at which interested applicants are expected to convene, and the hiring process and expectations are clearly communicated to all interested applicants. BCP staff members who are involved in hiring processes are expected to provide equal consideration to all applicants for a position, so long as they meet relevant criteria, through a process which is very sensitive to mitigating nepotism.

2.2.15 Relevant Laws and Regulations Related to Worker Rights (G3.11)

There are various legislations that have been enacted to protect workers in Zambia. Some of these laws include:

- **The Employment Act (CAP. 268):** An established guideline for employment of an employee in Zambia to be read together with the Minimum Wages and Conditions of Employment (General) Orders. It provides for dismissal of employees and the grievance procedures to be followed as well as basic employment rights relating to leave, working hours, maternity leave, minimum contents to be included in written contracts of service; retirement and redundancy benefits.
- **Minimum Wages and Conditions of Employment Act (CAP. 276) (Statutory Instruments (SI) No. 2 & 3) SIno. 46 of 2012 and the General Order, 2011:** Establishes minimum guidelines for remuneration and conditions of service for employees in Zambia setting out a minimum wage for specific sectors, basic rights and conditions of service.
- **Industrial and Labour Relations Act (CAP. 269):** Establishes guidelines on employee and employer organisations and the process of collective bargaining and dispute resolution. It provides for the formation of trade unions and employers representative organisations and recognises the right to strike under specified circumstances. The guidelines also provide procedures for the settlement of disputes, strikes and lockouts as well as the use of the Industrial Relations court at which employees can have their grievances and cases heard and determined.

BCP has a professional human resources and administration management team in place in Zambia that is familiar with Zambian labour laws and ensures strict compliance. The project meets or exceeds all applicable laws and regulations covering workers' rights.

BCP meets and exceeds urban minimum wage requirements for project area staff, many of whom are from the local community. BCP has also introduced benefits such as:

- Staff rations;
- “Healthy Green” allowance to ensure staff are able to procure healthy vegetables and protein for lunch;
- Private pension fund schemes to assist with retirement needs (to complement the State NAPSA pension fund);
- Inspiration fund to assist staff with career development;
- An emergency fund to assist staff during times of critical hardships;
- Additional private health cover for all staff and families, in addition to statutory contributions to the State NHIF health fund.

2.2.16 Occupational Safety Assessment (G3.12)

The Human Resources Manual contains a section concerning Occupational Health and Safety. All new BCP employees receive an induction that includes HR orientation (Environmental and social, and Human Rights Policy communication). It is expected that all BCP employees are aware of their rights and responsibilities. BCP continues to equip members with safety clothing, gloves and boots, and invest in additional training, such as first aid training, to ensure safe production practices.

BCP has developed an Environmental and Social Management System (ESMS) and continues to improve, implement and update it across all projects and companywide. In addition, BCP under ESMS has developed a serious incident reporting procedure that indicates possible cause to prevent recurrence. BCP tracks HSE incidents (Health, Safety and Environment; see Appendix L) in order to identify ways to improve the control environment to reduce risks on the job, and the ESMS department conducts annual HSE inspections at all BCP operation sites.

Prior to participating in any company supported activity, it is required that participants openly discuss any potential risks involved in the activity. Staff members are also expected to communicate any potential risks involved, and to provide participants with the necessary Personal Protective Equipment (PPE), such as work gloves or safety boots.

Participants in BCP-supported projects are explicitly instructed not to take unnecessary risks with their personal safety. BCP staff members are expected to provide regular oversight of project activities, and to communicate any concerns about safety to BCP’s management team.

Community Scouts (CS) receive training in firearm safety and arrest techniques from Government approved trainers. Their intimate knowledge of animal behaviour, having all grown up in the bush and received several training from external experts, ensures that all scouts are aware of how to avoid dangerous situations with wild animals. Refresher training for biodiversity is ongoing to ensure a deep understanding of the wildlife species in and around the project area. In the unfortunate case of a crisis, BCP has reaction plans in place. Additionally, all teams have access to transport and communication. All scouts are equipped with suitable boots and protective uniforms. Scouts have also been issued with strict official rules of engagement procedures to minimize risk and exposure to armed poachers.

Vehicle accidents are a risk to worker safety. Risks are minimized through maintaining a high standard of vehicle maintenance, and ensuring that drivers are well-trained and adequately monitored. Drivers' behaviour is tracked and reported in order to ensure correct action is taken to a conducive safe driving environment. BCP has regular ongoing intensive vehicle training sessions for all drivers of vehicles facilitated by qualified instructors. The training involved intensive road and off road driver training, defensive driving and vehicle recovery. BCP drivers sign a comprehensive vehicle policy, which strictly outlines company vehicle policy, intoxicants, limits of geographical use, speed limits set internally and an unauthorized passenger policy. In spite of the inherent risks of vehicle accidents in Zambia and on rural dirt roads, BCP is confident that the measures in place reduce this risk.

2.3 Management Capacity

2.3.1 Required Technical Skills (G4.2)

Key technical skills required to implement the project successfully include:

- Community engagement skills, including: community mobilisation, consultation, PLA/PRA techniques, sensitization, conflict-resolution, and awareness/respect for local practices;
- Biodiversity assessment and monitoring skills, including proficiency in landscape ecology, conservation biology, statistics and GIS;
- Carbon measurements and monitoring skills, including proficiency in forest ecology, forest dynamics, Remote Sensing, GIS, Statistics and script coding, with the ability to adapt to and implement new/revised methodologies;
- Government engagement skills, including government engagement and consultation, knowledge of and respect for local laws, ability to ensure compliance;
- Business development, sales and marketing skills, relevant to the ability of the project to develop a sustainable financial model, based on the generation and sale of verified credits, to fund ongoing project activities, and to sell offsets at the highest price;
- Land and Forest Management Planning;
- Law enforcement, patrol planning;
- Aviation Air wing support skills: BCP's model uses aviation primarily to support flying FPIC, as well as monthly monitoring. Given the scale of LCFP, Air Wing support is a required technical skill;
- Strategic Fire management skills;
- Finance and Administration management skills;

The maintenance of these required skills is ensured by a continuous recruitment of leading experts in their respective fields, as well as through training opportunities to staff (see Appendix A). BCP also works with consultants and has built strong partnerships with leading organisations such as Lion Landscapes (UK research and conservation organisation collaborating on BCP's biodiversity monitoring plans) and The Landscapes and Livelihoods Group (collaborating on BCP's Environmental and Social Risk Management Plan).

2.3.2 Management Team Experience (G4.2)

The LCFP management team has acquired significant and unique skills and experience relevant to the implementation of large-scale land management and carbon projects in Africa (Appendix A).

Since 2012, BCP's management team has been responsible for the development and implementation of the Lower Zambezi REDD+ Project (LZRP), which was the first verified REDD+ activity in Zambia. The LZRP was also the first REDD+ project in Africa, and the third in the world, to achieve gold-level validation against all three criteria of the Climate, Community and Biodiversity (CCB) standard. The BCP management team has continued to successfully manage the LZRP since 2012, and has achieved yearly successful VCS verifications since 2013: the first REDD+ project in Africa to obtain this milestone and the 2nd in the world. In 2022, BCP had successfully passed 16 validations and verifications under VCS and CCB standards when considering both REDD+ projects, LCFP and LZRP.

BCP's Founder, Dr. Hassan Sachedina was the Vice-President of Wildlife Works, when the US firm verified the world's first REDD+ project using VCS methodology to a project in Kenya. Since then, Dr. Sachedina has helped to set up 6 VCS verified REDD+ projects in Kenya and DRC and 2 projects in Zambia. Together, these projects cover an estimated 2.1 million ha and benefit over 500,000 people.

The BCP management team has over 30 years of cumulative dedicated experience focused on the implementation of REDD+ in Zambia, conservation, and carbon-related activities. BCP team of core managers includes competent specialists in conservation, ecology, carbon, community engagement, business administration, aviation, Air Wing, sales and marketing. BCP is supported by implementing partners from Government institutions with relevant authority and skills to assist in forest and biodiversity management and community engagement.

2.3.3 Project Management Partnerships/Team Development (G4.2)

Below is a list of key partners who provide BCP with critical expertise to Community and Biodiversity work:

1. Livelihoods Partners: BCP partnered with honey production firms, Bee Sweet until 2020, followed by Nature's Nectar, whose focus is on selling honey to the international market. The aim is to use an innovative approach of outsourcing hive management to farmers through beekeeping mentors who distribute and help farmers manage their hives. This way BCP ensures the transfer of technical knowledge of honey production to communities for sustainability.

Another common challenge addressed under this project is access to markets for honey producers. The chosen honey partner for this project is able to guarantee farmers a market at a good price for the honey they produce. BCP also consults with The Landscapes and Livelihoods Group (TLLG) who is helping to develop an Environmental and Social Risk Management system, Environmental Monitoring and Management Plan and other components related to improving BCP's impact on community livelihood, through the use of practical and community-oriented approaches following international standards. TLLG also assists BCP to comply with the International Finance Corporation (IFC) Policy and Performance Standards.

2. Biodiversity Partners: From the start of the project, BCP has partnered with Lion Landscapes on the development and implementation of biodiversity monitoring plans. The collaboration agreement MoU was signed in October of 2014 and renewed on February 10, 2020. This collaboration resulted in successful

grant applications notably to the Darwin Initiative and Lion Recovery Fund. BCP also signed an MoU with Panthera on August 18, 2021 an organisation specialised on felids' monitoring and conservation. Panthera provides support for the Spatial Monitoring and Reporting Tool (SMART). In 2022, BCP signed two new MoUs with Conservation organizations, namely BirdWatch Zambia for the monitoring of vultures and other key bird species, and Mabula Ground Hornbill Project for the monitoring of the Southern Ground-Hornbill. Other organisations provide support, either through formal agreements such as with Biodiversity Research Institute, or informally such as the Zambian Carnivores Program which shares information on African wild dogs.

3. Other Implementing Partners: BCP has signed MoU's with Peace Parks Foundation, The Nature Conservancy, and Game Rangers International for the implementation of new REDD+ activities.

4. Government: Forestry Department (FD) and Department of National Parks and Wildlife (DNPW) are the authorities and regulators when it comes to forest legislation and wildlife laws. These Departments are well versed and skilled in community engagement and FPIC, as community benefit from natural resources is a key part of their mandate. Government Officials, especially at District level, in varied departments are specialists and familiar with local customs and geography. On all levels, the Government is a key partner of this project.

Another key partner includes the Ministry of Agriculture, through the District Agriculture Coordinating Officers referred to as DACOs. The DACOs provide training to lead farmers in Conservation Farming techniques. The government of Zambia has invested in training their agricultural extension agents in basic Conservation Farming techniques that help farmers intensify their agricultural activities, build resilience to climate change and improve productivity per hectare. Under the LCFP, BCP is working under an existing MoU with each District Agriculture office.

5. Community Institutions: Community institutions are critical partners as they are the gateway to communities, and entrusted, by law, to represent and serve their communities.

2.3.4 Financial Health of Implementing Organization (s) (G4.3)

The financial stability of BCP is presented as part of the non-permanence risk assessment (Appendix L) and financial analyses for both BCP REDD projects.

2.3.5 Avoidance of Corruption and Unethical Behaviour (G4.3)

BCP adheres to local laws and international standards that are further reinforced by internal policies preventing corruption and unethical behaviour. There is a zero tolerance policy toward corruption, bribery, embezzlement, extortion or collusion. BCP's anticorruption policy is found in the BCP Standard, as well as the company's 'Roots' - an overarching set of company values developed by BCP staff in a consultative manner.

2.3.6 Commercially Sensitive Information (Rules 3.5.13 – 3.5.14)

There is no commercially sensitive information that has been excluded.

2.4 Legal Status and Property Rights

2.4.1 Recognition of Property Rights (G5.1)

The right of property use is expressed to the Chiefdom Community Forest Management Groups (CFMG) as agreements between BCP and the community structures in place. In regards to GHG emission reduction, the ability to claim that the project will cause reduction is expressed through project activities and monitoring arrangements in the Community Forest Management Partnership Agreements (CFMPA). For private ranches a contractual arrangement is proposed and no transfer of property rights occurs (See Appendix K).

The Forest Act 2015 has been instrumental in developing guidelines that secure statutory rights such as the Forest Carbon Stock Management guidelines and the Statutory Instrument NO. 11 OF 2018 [The Forests Act, 2015 (Act No. 4 of 2015)] The Forests (Community Forest Management) Regulations, 2018 that will guide and secure statutory rights (see section 2.4.3 for details on the regulatory property rights recognition and protection).

Property rights are recognized under The Forests (Community Forest Management) Regulations (2018) where a local authority, for forests within the jurisdiction, identify, support and encourage local communities to apply to the Director for control, use and management of areas of forest. This is for the purpose of social, cultural and economic needs.

A CFMG may apply to the Director for recognition in Form I set out in the schedule (section 6 (1)). The community forest agreement shall specify the right to issue community permits and collect revenue for products specified in the community forest management plan (11 (4)).

Where a CFMG intends to assign some or all of its rights under a Community Forestry Agreement to another person or group, the group shall apply to the Director for the transfer of rights in Form VII. The person or group assigned the rights will be subjected to the requirements of the approved community forest agreement and additional conditions specified in the Schedule. The CFMG and person/group assigned rights shall specify the benefits sharing arrangement.

2.4.2 Free, Prior and Informed Consent (G5.2)

The development of REDD+ projects supported under LCFP adheres to strict standards for ensuring adequate stakeholder consultation and engagement take place. This is in line with international standards for ensuring Free, Prior and Informed Consent (FPIC).

In 2014, BCP facilitated a series of workshops involving numerous representatives from Government and civil society, to collaboratively develop a Standard Operating Procedure (SOP) for FPIC in Zambia. This was finalised in early 2015 and has been updated on a regular basis. BCP is currently working with Version 8 (Appendix J). From project inception, BCP has ensured compliance with the SOP terms, throughout all FPIC activities and subsequent stakeholder engagements.

Stakeholder consultation and engagement, in line with the principles of FPIC, were a central component of BCP's REDD+ viability assessment process in 2015 and 2016 through to the point of receiving community consent letters for REDD+ implementation.

REDD+ project stakeholders supported under the LCFP include Government partners and community members from 17 Chiefdoms. These community members are represented by CFMGs and VAGs, and fall under the jurisdiction of the Royal Establishments (Chiefs/Chieftness, supported by traditional leaders such as Headmen and Indunas). BCP possesses a full set of documentation, confirming significant community consultations and engagements took place, over a period of more than 2 years, in line with the principles of FPIC, in all 17 partner Chiefdoms for the LCFP – prior to receiving any consent letters for REDD+ implementation, from these same community stakeholders.

In 2016, BCP received signed consent letters for REDD+ implementation from all initial partner Chiefdom community stakeholders and implementing partner CFMGs, as well as VAGs local-level institution representatives. These representatives are recognized to have the authority to make commitments on behalf of the community, according to local laws, customs and traditions (see Appendix K).

These community consent letters to REDD+ implementation represent the end of the viability assessment process. BCP then signs two subsequent agreements with community stakeholders, further documenting specific commitments with regards to forest protection activities, and community benefit sharing. As of 2017, BCP have additionally signed in-depth Community Forest Management Agreements (CMFAs) that function as Participatory Forest Management Plans.

BCP possesses a full set of supporting documentation to confirm that FPIC processes took place, according to the SOP and international standards, including the Social Monitoring Surveys (Appendix E). These documents collectively shows:

- Local communities have been widely and appropriately consulted, prior to project implementation;
- Local communities are aware of, and freely consenting to the terms and boundaries of the REDD+ projects;
- Local community stakeholders were involved in participatory forest selection and project design, through a series of PRA/PLA activities in each Chiefdom, which were led by trained facilitators, and fully documented;
- Stakeholders were provided with accurate information concerning the activities, location(s) and size(s) of the planned project. The nature of REDD+ activities is supported with relevant timelines, scope, personnel involved, and reasons/purposes for REDD+ activities to be undertaken. Stakeholder feedback was solicited, and used to identify terms of proposed the REDD+ projects – as stakeholders, identified forest areas for protection, identified reasons to implement REDD+, and identified deforestation mitigation activities to be supported– in addition to reviewing existing literature and studies relevant.

Additionally, BCP's set of FPIC records for LCFP demonstrate the following steps took place, in each Chiefdom:

Step 1: Initial “desk review” research to identify potentially viable REDD+ areas according to the liability criteria concerning forest composition, project locations, potential stakeholders and project impacts/risks and relevant data/analysis/studies of previous attempts.

Step 2: Initial consultations with key (high level) stakeholders, including district level workshops /meetings, and/or initial consultations with traditional leaders and community institutions. At, or following, these workshops, inputs from key stakeholders were sought with regards to stakeholder identification, initial

proposed forest areas for REDD+ and high-level stakeholder permission for the viability assessment processes to proceed (viability proposed forest area assessments and FPIC consultations).

Step 3: Participatory forest selection and stakeholder consultations, in line with FPIC, using PRA/PLA techniques. Including community consultation meetings at Chiefdom, VAG and village levels, as well as individual stakeholder consultations to ensure widespread information dissemination.

Step 4: Follow-up meetings were held at Chiefdom level, to confirm community stakeholder consent. Members of the wider community were invited to attend and provide feedback, and/or ask questions, before relevant authorities (VAG and CFMG representatives and traditional leaders) were approved by the community, and Chief/Chieftness - as the ultimate authority, to consent to the terms of the proposed project. These meetings resulted in community consent letters for REDD+ implementation, signed by all relevant authorities, on behalf of the community.

Community consultations took place in line with a standard “curriculum” that was developed, as outlined in the FPIC SOP. Representatives from BCP and implementing partners, including Government, received technical training in REDD+, FPIC and PRA/PLA, prior to undertaking community sensitizations. Locally appropriate informational materials were developed, and refined (based on experience) to communicate key messages about REDD+ to local community stakeholders. These included visual aids, and materials that were translated into local languages, to accommodate the variety of languages spoken by LCFP stakeholders, as well as high levels of illiteracy. Trained facilitators provided information and led meetings/workshops/consultations in local languages. All of the informational materials are available as supporting documents, upon request.

As a result of competing FPIC processes, BCP does not undertake any activities that are encroaching on any form of land; REDD+ activities of the LCFP are only being implemented in areas where the community and government stakeholders to the project are knowingly and willingly participating. BCP has complied with relevant local and international standards and policies for REDD+ implementation, and the activities of this project are compliant with, and will continue to comply with and be protected by relevant legislation in place (as outlined in Section 2.4).

Throughout the community consultation and feedback processes, as part of the REDD+ viability assessment process, community stakeholders have consistently confirmed that the REDD+ activities of the LCFP are being implemented on areas of land that are free of encroachment. The REDD+ activities of the LCFP will not cause any evictions or relocations of established communities. In the event of any individual relocations – for example, if it is found that an individual or household has recently encroached into an area proposed for REDD+ implementation – relocation arrangements will be agreed by all relevant stakeholders, through agreed / appropriate community institutions, and will take place according to mutual agreement. As the Conservation Fees Guidelines Agreements signed between BCP and local community stakeholders document, an agreed grievance resolution mechanism is in place, and will be supported by BCP through the LCFP, to ensure that any grievances, concerns or questions may be reported and appropriately responded to.

2.4.3 Property Right Protection (G5.3)

Zambia has one of the most enabling legislations in the world for empowering and protecting community rights through the Forests Act of 2015 and 2018 CFM SI, which enables communities to be issued major forest produce rights.

Forest user rights are granted to the community by the Director of Forestry as in section 31 (1) of the 2015 Forest Act which states “A Community Forest Management Group (CFMG) recognised under section thirty may apply to the Director, in the prescribed manner and form, to enter into a community forestry agreement with the Department in respect of an area or forest for which the CFMG is formed. (2) The Director shall, in considering an application under subsection (1), after consultation with the Chief and the local authority in respect of an area or forest for which the CFMG is formed.”

Referring to the SI NO. 11 OF 2018 on Community Forest Management Regulations, a local authority, for forests within the jurisdiction, identify, support and encourage local communities to apply to the Director for control, use and management of forest areas for purposes of social, cultural and economic needs.

Under section 4, (1) Subject to these Regulations, community forest management may be applied in - (a) open areas; b) local Forests; and (c) Game Management Areas.

Under section 6, (1) A CFMG may apply to the Director for recognition in Form I. (2) An application for recognition of a CFMG shall require; (a) prior consultations with local users and rightful holders of the proposed forest; and (b) the consent of local traditional leaders.

Under section 3, An application for recognition shall be accompanied by: (a) a sketch map of the proposed location and area to be established as a community forest; (b) CFMG constitution including list of the elected representatives of the group; (c) consent of the Chief of the area to recognise the applicants as a CFMG through endorsement of the application or map signifying such consent; and (d) a statement of intent for the proposed area indicating a balance between forest protection, management, development, utilisation and forest enterprise development in Form I set out in the Schedule.

Under section 8, (1) The Director shall, where the applicants meet the requirements referred to in regulation 6, recognise a group as a Community Management Group. Section 9 (1) states that the community forest management area shall be for the exclusive use of the local community represented by the CFMG unless an application for recognition is rejected by the Director.

Therefore, under section 3, The Community Management Groups shall: (a) protect and restrict access to the community forest area by persons not part of the community; (b) control restricted activities within the identified community forest area; and (c) report forestry related offences to the forest officer. This is a guarantee that project activities do not lead to involuntary removal or relocation of property rights holders from their lands or territories, and does not force rights holders to relocate activities important to their culture or livelihood. The Community Forest Management Agreement (CFMA) provides rights, duties and guides for both parties.

Under Section 10, (1) A recognised CFMG shall carry out field appraisals and socio economic survey in the targeted community forest in order to conduct an assessment on the potential user rights by the community and other users of the targeted community forest.

Under section 11, (1) A recognised CFMG under regulation 8 shall, within 90 days of recognition, enter into a Community Forest Agreement in Form IV set out in the Schedule. Further, under section 11, (3) An application to enter into a CFMA shall be accompanied by; (a) a community forest map showing the boundary of the community forest that has been consented to or signed by representatives of concerned neighbouring communities, Ward Development Committees, and the Chiefs within the area proposed for community forest management; and (b) a Community Forest Management Plan (CFMP) which is in

accordance with sustainable forest management principles that includes a set of community rules and measures that support proper management of the community forest.

Under section 11, (4) The Community Forest Agreement shall specify the right to issue community permits and collect revenue for products specified in the CFMP.

Under section 16, (1) The CFMGs shall assist the Director to control restricted activities within an established community forest area.

Under section 17, (1) Where a CFMG intends to assign some or all of its rights under a Community Forest Agreement to another person or group, the group shall apply to the Director for the transfer of rights in Form VII set out in the Schedule. Under section 17, (2) the person or group being assigned the rights shall be subjected to requirements of the approved Community Forest Agreement and additional conditions specified in the Schedule. Under section 17, (3) the CFMG and people to be assigned rights shall specify the benefits sharing arrangements.

BCP has facilitated multiple CFMGs within Zambia to the rights transfer stage with the Government.

Under LCFP, relocations are a preserve of the CFMG in consultation with the traditional authority and the local authority. A similar situation applies to the private properties, as there is a contractual agreement in place and no rights are violated.

2.4.4 Identification of Illegal Activity (G5.4)

Three illegal activities could potentially affect the project impacts:

- Selective wood extraction for timber and charcoal production: Measures in place to reduce these activities include development of land use plans, building law enforcement capacity jointly with Government and community partners, and alternative livelihoods such as beekeeping and Eco-Charcoal.
- Illegal land allocation by indunas and village headmen: May cause encroachment in Community Forests, and undermine the Community Forest Management Partnership Agreement (CFMPA). Measures to reduce these activities include empowering the CFMG and the Community to hold their local leaders accountable under the Forests (Community Forest Management) Regulations, SI No. 11 of 2018. The community value creation package carried out by the project will also be key to ensuring households at community level benefit and to engender the principle of “social fencing”: habitat protection carried out by local people without the use of actual physical fencing.
- Illegal Wildlife Trade: Measures needed and designed to reduce these activities include on-ground surveillance and conservation activities.

These illegal activities are also countered by regular aerial and ground monitoring of the project areas.

2.4.5 Ongoing Disputes (G5.5)

No ongoing disputes were registered up to the close of this monitoring period.

2.4.6 National and Local Laws (G5.6)

Relevant laws for the project activities are listed below and available in Appendix K. These Laws have not been altered and no new laws/regulations have been put into effect since the last validation.

Environmental Management Act No. 12 of 2011

The Act provides for the establishment of the Zambia Environment Management Agency (ZEMA) whose main functions constitute the protection of the environment and control of pollution in particular, so as to provide for the health and welfare of persons, animals, plants and the environment in general.

The ZEMA has no objection against the implementation of the LCFP REDD+ Project and has exempted it from any Environmental Impact Assessment.

Mines and Minerals Development Act (No. 7 of 2008)

The act is relevant to the project as activities related to large scale mining might take place within the project area. The act provides for regulations concerning environmental protection during construction, operation and closure of mine sites.

Zambia Wildlife Act 2015

The Act provides a legal framework for the Zambia Wildlife Authority to establish, control and manage National Parks. Enabling conservation and protection of wildlife and National Parks, as well as establishing Game Management Areas, licenced hunting, control of trophies and bushfires.

Water Resources Management Act

The Act was established in 2011 and repealed the Water Act of 1949. It provides control, ownership and use of water. The Act provides for the management, development, conservation, protection and prevention of pollution of water resources and its ecosystem. It requires issuance of water licences and permits and rules relating to construction of dams on existing rivers. In addition, the Act provides a right of easement over land under specified circumstances, including the right of access to any piece of land contiguous to the water which is subject to a permit. Note there is a difference between public and private water.

Forests Act, 2015

The Act provides for the establishment and management of: National and local forest conservation; Forestry Commission and participation of local communities; NGO's and traditional institutions for sustainable management of forest ecosystems and biodiversity; Licensing and sales of forested land; Declaration of local forests, forest plantations or open areas to be a Joint Forest Management Areas

The implementation of various international conventions relating to the environment is also included. It is also important to note the President can declare any land a National Forest and may, by compulsory acquisition acquire any land for the purpose of a National Forest.

Zambia Tourism and Hospitality Act No. 24 of 2007

Enacted in 2007, the Act provides for the control of tourism enterprises. The Act, though making no direct reference to environmental protection, provides for appeals against authorization of projects which could negatively affect Zambian tourism.

National Heritage Conservation Commission Act No. 23 of 1989

Enacted in 1989, the Act provides for the conservation of ancient, cultural and natural heritage, relics and other objects of aesthetic, historical, prehistoric, archaeological or scientific interest.

The Employment Act (CAP. 268):

Establishes guidelines for employment of an employee in Zambia to be read together with the Minimum Wages and Conditions of Employment (General) Orders. Guidelines for dismissal of employees and grievance procedures are to be followed as well as basic employment rights relating to leave, working hours, maternity leave, minimum contents to be included in written contracts of service; retirement and redundancy benefits.

Industrial and Labour Relations Act (CAP. 269):

Establishes guidelines for employee and employer organisations regarding: The process of collective bargaining and dispute resolution; The formation of trade unions and employer representative organizations; Recognizes the right to strike under specified circumstances; The settlement of disputes, strikes and lockouts; The use of the Industrial Relations court where employees can have their grievances and cases heard and determined.

The Land Acquisition Act, Chapter 189 of the Laws of Zambia

The Act determines provisions for the compulsory acquisition of land. The acting President can compulsorily acquire land that is desirable or expedient in the interest of the Republic, subject to various obligations which include the requirement to give notice of intention and compensation payments. Adequate compensations shall be paid and any disputes shall be determined by the courts.

The Lands Act, Chapter 184 of the Laws of Zambia

This Act enables the continuation of leasehold tenure and vesting of land through statutory recognition and acceptance by the President. It also enables conversion of customary tenure to leasehold tenure. The Act explains that a person shall not without lawful authority, continue to occupy vacant land and any person who does shall be evicted.

Note that customary land existed in Zambia before colonialism – under the unwritten indigenous laws of tribes. Customary law can be used in settlement disputes involving members of tribes and is vital in the settlement of land disputes held under customary tenure.

2.4.7 Project Benefit Crediting (G5.9)

The LCFP is registered to the VCS Program under the project ID 1775. The LCFP is not subject to binding emissions limits or emissions trading programs, and at this stage is not receiving or seeking to receive another form of social or environmental credit.

3 CLIMATE

3.1 Net Positive Climate Impacts

3.1.1 Net Impact (CL2.2, CL3.1, CL3.3)

A summary of net GHG removals and reductions are provided in Table 10. Emissions were calculated using the Approved VCS Methodology VM0009, v3.0 (Appendix B).

Table 10. Summary of net GHG removals and deductions for the current monitoring period by vintage year

Year	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals (tCO ₂ e)
2015	570,024	84,506	0	485,518
2016	1,088,751	126,758	0	961,993
2017	1,317,124	126,412	0	1,190,712
2018	1,460,802	126,412	0	1,334,390
2019	1, 570,988	75,975	0	1,495,023
2020	1,743,915	150,080	0	1,593,835
2021	1,970,329	41,326	0	1,929,003
2022	2,374,072	66,090	0	2,307,982
Total	12,096,015	797,559	0	11,298,456

3.2 Offsite Climate Impacts (Leakage)

3.2.1 Leakage Mitigation (CL3.2)

The primary leakage mitigation strategies of the project are improving agricultural techniques with conservation farming and agroforestry and promoting alternative livelihoods that don't depend on forest clearing such as beekeeping. Based on the experience of BCP's other REDD+ project in Lusaka Province, which employed similar leakage mitigation activities and has not experienced any leakage in 10 years, it is assumed that these activities will also be effective at preventing leakage from the LCFP project area. For additional information refer to Appendix B Monitoring Report 1 to 5.

3.3 Climate Impact Monitoring

3.3.1 Climate Impact Monitoring Results (CL4.1)

Table 11. Description of expected short-term and long-term impacts of project activities

Monitoring activity	Timing	Methods	Results
Trespassing and boundary	Monthly patrols backed up with aerial surveys at least once per dry	Scouts patrol the project are on an ongoing basis, it is their	78 BCP-supported scouts enrolled to monitor and support the

Monitoring activity	Timing	Methods	Results
transgressions	season.	core duty	preservation of the forest. Scout patrol reports can be found in Appendix G
Leakage monitoring	Annually	Forest Inventory team re-measuring 100% of leakage plots each year as per protocol.	Yearly leakage monitoring results can be found in VCS monitoring reports (Appendix B)
Plot measurements	Annually	Forest Inventory Team following VM0009, re-measuring 20% of plots in project accounting and project proxy areas as per protocol. Since all the plots (100%) were measured a year prior to project validation (i.e. 2018), the 20% per year rotation started in 2019. An additional 40% were measured in 2020 and 2022. Therefore, a 100% rotation will be completed in 2023.	Additional information can be found in VCS monitoring reports (Appendix B)
Disturbance monitoring	Once a year using satellite imagery from April-May	Remote sensing using Sentinel-2 or Landsat imagery. Followed up with aerial surveys and, or scout patrols.	A total of 3492.71 ha of forest were lost between 2017 – 2022. Results of disturbance mapping in the project area can be found in Appendix B and C

For the monitoring of community and biodiversity adaptation to climate change, refer to section 4 and 5 respectively, in addition to the section 3.4.2 below.

3.3.2 Dissemination of Monitoring Plan and Results (CL4.2)

All verification documentation is made available to communities and other stakeholders. Project activities are discussed at community consultation meetings, reported in newsletters and updates posted on the BCP website.

The monitoring plans for this project are shared as a publicly available appendix to this project document on the VCS / CCB project databases (<https://www.vcsprojectdatabase.org/>). Verified monitoring reports are also being made publicly available through the same online database.

The Results of the forest monitoring plan are disseminated to the communities during the meetings held at the village level. Leaflets summarising the monitoring plan and results, translated in local languages, have been distributed to each community involved in the project (see Appendix J).

3.4 Optional Criterion: Climate Change Adaptation Benefits

3.4.1 Activities and/or Processes Implemented for Adaptation (GL1.3)

Table 12. Summary of monitoring activities

Climate change impact	Causal model	Project adaptation activities/ measures
Reduced yield quality and quantity due to: - Increase in extreme rainfall events - Increase in mean annual temperature - Increased risks of droughts and floods³⁰.	Subsistence agriculture is particularly dependent on seasonal rainfall, which is predicted to change over the next 30 years.	- Income diversification - Improved agriculture techniques - Conservation farming/Climate Smart Agriculture - Beekeeping - Improved production - Improved access to market
	Due to the projected increases in temperature and the variable nature of precipitation in the future, crop failure threatens both food security as well as biological diversity.	
	Education will provide opportunities for climate change adaptation for community members that go beyond subsistence agriculture.	Improvement of school facilities through refurbishment and construction of schools and teachers' houses, refurbishment/construction of boreholes at proximity of the schools
- Water shortage during the dry season, resulting in habitat compression and degradation around remaining water sources. - Increased risk of late season wild fires due to longer and hotter dry	Agents within the project zone are more likely to exploit natural resources through poaching and charcoaling activities if their crops fail as a result of climate change (increase in temperature and erratic rainfall patterns).	Securing long term conservation of a vast wildlife corridor along a north-south rainfall gradient and ensuring access of wildlife to critical water sources through avoided deforestation and degradation and reduced hunting pressure. Reducing the risk of destructive wildfires by implementing and

³⁰ McSweeney, C., M. New, and G. Lizcano. 'UNDP Climate Change Country Profiles: Zambia', 2010.

seasons		promoting early dry fire management systems.
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The project provides significant support to communities and biodiversity to adapt to anticipated climate change impacts and risks.

The Zambian climate is heavily influenced by El Niño-Southern Oscillation (warm phase) related events that tend to bring unusually high rainfall in the north with the opposite conditions prevalent in the south. Conversely, La Niña (cold phase) events result in dry conditions in the north and wetter conditions in the south. With the uncertainty associated with predicting these events it is difficult to proactively negate the potential impacts. An example of the impacts these phenomena have is the 1997-1998 El Niño event, where abnormally high rainfall in the north resulted in persistent flooding while near drought conditions were experienced in the south. Similarly, a drought in 2004-2005 resulted in major agricultural losses throughout the country.

Projected climate change impacts will exacerbate current problems related to poverty and vulnerability. Agents within the project area are heavily dependent on the natural environment for their day-to-day livelihoods. Subsistence farming is the general agricultural practice in the area; most farmers produce just enough to feed themselves and their families and supplement their income by keeping chickens and/or goats, with some practising charcoal production as a means of earning additional income. Climate change impacts will likely have a negative effect on the livelihoods of these people, as their resilience to climate change is very low. An example of this is the dry land farming commonly practised—this form of subsistence agriculture is particularly dependent on seasonal rainfall, which is predicted to change over the next 30 years.

3.4.2 Adaptation Monitoring (GL1.4)

1) Community Adaptation

Projected changes to the climate suggest that the dry season may become longer with reduced rainfall in the September-November period, exacerbated by an increase in extreme rainfall events in the December-February period. Mean annual rainfall is projected to remain largely unchanged, however, the increase in the extreme rainfall events and the increase in mean annual temperature suggest that risks associated with droughts and or floods should be planned for using flexible, reactive and anticipatory management strategies. Additionally, while projections may be uncertain, in a semi-arid region such as the LCFP any social development project should account for the natural variability inherent in these systems.

Due to the projected increases in temperature and the variable nature of precipitation in the future, crop failure threatens both food security as well as biological diversity. Agents within the project zone are more likely to exploit natural resources through poaching and charcoaling activities if their crops fail.

The primary project activities designed and implemented to deliver the required outcomes related to climate change adaptation, reducing vulnerability and increasing resilience of local communities include,

- Climate smart agriculture (CSA) extension support to lead farmers and follower farmers where agents are encouraged to practise conservation farming methods such as minimum tillage and moisture conservation practices. During the monitoring period more than 408.5 ha were cultivated

under climate smart agricultural practices. As a result of the implementation of CSA practices, crop yield experienced an increase of 80-175% for maize grain compared to the baseline (Appendix H).

- Supporting the production and sale of locally harvested honey as an alternative income. The project proponent supports honey production projects within these communities through training, provision of hives, regular monitoring and guaranteeing access to both local and regional markets. This activity diversifies household incomes and provides a buffer in case of climate induced agricultural shocks. During the monitoring period the number of households that collected honey for sale and/or consumption or bought from others increased from 8.3% in 2020 to 14.33% in 2022 (See Appendix I (01_IPS)).
- Improved horticulture farming. During the monitoring period over 172 lead farmers were trained in horticulture practices.
- Poultry and livestock support programs. During the year 2021 over 210 goats and 100 chickens were distributed among 70 beneficiaries in Kawala, Wazaza and Mudima VAGs) (see Appendix B (MR4) and J)
- Social services: Education creates opportunities for climate change adaptation to community members that goes beyond subsistence agriculture. As such, the project proponents work with CFMGs to ensure carbon funds are used to advance health, education and WASH projects. During the monitoring period more than 200 community and 150 livelihood projects respectively, were implemented across the project area. Further information can be found in section 4.3.1 and Appendix J.

2) Biodiversity Adaptation

The main biodiversity adaptation of the project is securing the long term conservation of a vast wildlife corridor along a north-south rainfall gradient and ensuring access of wildlife to critical water sources. The project also reduces the risk of destructive wildfires by implementing and promoting early dry season fire management systems.

During this monitoring period, a fire management plan including firebreaks and early burning have been implemented to reduce the risk of wildfires (Appendix G).

4 COMMUNITY

4.1 Net Positive Community Impacts

4.1.1 Community Impacts (CM2.1)

Table 13. Smallholder farmers benefiting from Conservation Farming/Climate Smart Agriculture

Community Group	Smallholder farmers: Lead farmers and follower farmers
Impact	Project Activity: Conservation Farming/Climate Smart Agriculture/horticulture Impact: climate change resilience, income level, health and diet
Type of Benefit/Cost/Risk	Actual impact benefiting communities
Change in Well-being	Type of change: higher diversity of crops, improved drought resilience, more diverse diet, improved health and income level Justification: The traditional form of subsistence agriculture is particularly dependent on seasonal rainfall, which is predicted to change over the next 30 years. Drought vulnerability will be reduced by the implementation of conservation agriculture methods such as minimum tillage and moisture conservation practices. Expected outcomes: • Increased production of high-value crops (soya beans, cowpeas, sunflower). • Improved income levels of rural households through the sale of surplus grains. • Enhancement of nutritional status of farm families, especially in children and women through increased consumption of pulses. • Confidence built amongst small-scale producers due to cultivation using sustainable methods and diversifying food production, and entering into small-scale agri-business practices. • Reduced dependence on agriculture subsidies and use of artificial fertilisers. • Enhanced knowledge base and skills of small-scale rural producers in modern production, processing and marketing practices. • Access to the local, regional and national markets ensured for small-scale rural producers through market linkages Magnitude of impact: During the monitoring period, 1386 lead farmers were trained in Climate Smart Agriculture (Appendix A). By the end of 2022, 408.5 ha were under such type of sustainable farming practices (Appendix H). As a result of smart agriculture technologies, farmers' mean maize grain yields still remained at the higher level (4 ton/ha) than the baseline level (1.8 ton/ha) (Appendix H). The 2022 data

	showed that average household income increased by 220% over the five-year period with agricultural production, alternative sources and forest-based products being the key sources of income for the households (see Appendix E). It has been demonstrated that revenues obtained from conservation farming surpass by far the levels of income generated under traditional farming practices Impact evaluation method: Empirical assessment of the annual yields quantity and quality average household income level.
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Table 14. Smallholder farmers benefiting from Beekeeping

Community Group	Smallholder farmers: beekeeper
Impact	Project Activity: Beekeeping Impact: climate change resilience, income level, health and diet
Type of Benefit/Cost/Risk	Actual impact benefiting communities
Change in Well-being	Type of change: improved income level, diverse diet, improved health Justification: Beekeeping is well suited for individuals and companies with limited resources for investment and it's potential to contribute to the reduction of rural poverty is important. As opposed to subsistence farming, beekeeping is less likely to be impacted by changes in rainfall pattern. Expected outcomes: Although honey hunting and beekeeping have been practised by almost all communities in Zambia, it has only recently been promoted as an alternative livelihood for people involved in charcoal production, illegal game hunting, and those that want an extra income. Additional outcomes include: • Increased production of natural honey. • Increased income levels of rural households through honey sales. • Reduction in tree cutting for traditional honey hunting. • improved survival of bees threatened by use of chemicals and pesticides for crops. • Enhanced knowledge base and skills regarding modern honey production, processing and marketing practices. • Confidence built amongst honey producers due to a guaranteed market. Magnitude of impact: • Partnership with Nature's Nectar, 2020 to present (Appendix H): - 225 honey producers- lead farmers (Mentors) trained - 7,998 beehives distributed - 1,089 beekeepers registered (2022 data not included)

	- 6,020 kg of honey harvested - 135,018 ZMW generated from honey sale (7,900 USD) Impact evaluation method: Number of beehives distributed, number of beekeepers trained, quantity of honey harvested and total revenue generated through the sale of honey.
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Table 15. Smallholder farmers benefiting from infrastructure development and social service delivery

Community Group	All community members
Impact	Project Activity: infrastructure development and social service delivery Impact: education and health
Type of Benefit/Cost/Risk	Actual impact benefiting communities
Change in Well-being	Type of change: Improved access to education, clean water, and health care services Justification: Education will provide opportunities for climate change adaptation for community members that go beyond subsistence agriculture. If BCP can help communities make long term investments for development and demonstrate that those investments are intrinsically linked to the successful long term conservation of forests, then livelihoods in those communities will improve demonstrably and be sustainable, while natural forests will remain intact. Magnitude of impact: The summary of infrastructure development and social service projects implemented by the LCFP is presented in Appendix J. During the monitoring period more than 200 projects have been completed. Impact evaluation method: Number of projects completed and number of project beneficiaries

4.1.2 Negative Community Impact Mitigation (CM2.2)

Negative impacts revolve around the possibility of conflicts arising due to perceptions of unfair benefit sharing. Benefit distribution has been carried out in a transparent and fair manner that ensures the participation of all VAGs. Financial expenditures by the CFMG and VAGs are prominently shared with constituents.

Community members who depended on illegal activities to support their livelihoods (i.e. poaching, illegal charcoal, cutting trees for construction materials) will experience short term negative impacts as the LCFP project seeks to curb these activities and improve law enforcement. The measures to mitigate the negative impacts the project may cause on livelihoods, are driven by a combination of job creation, and livelihood alternatives. At implementation, the project created approximately 108 community jobs, mainly for Community Scouts (CS). These CS are hired through BCP support via CFMG contracts with the role to protect community resources, including wildlife habitat. The income generated goes directly into the Chieftdom, and all community members have an equal opportunity to apply for CS jobs. At the end of this

monitoring period, over a hundred community members are on full-time jobs with BCP. As most development activities benefit the community, it is expected that the long-term well-being of poachers and charcoal producers' households will improve.

The HCVs present in the Project Zone are now benefiting from formal protection from human activities that can hinder the value of these HCVs (see section 4.1.4 below).

4.1.3 Net Positive Community Well-Being (CM2.3, GL1.4)

In addition to the specific community livelihood activities listed above in section 4.1.1, it should be noted that these impacts take place within the wider framework of the effective development and implementation of community-based REDD+ activities, under the LCFP. As such, additional impacts include:

- Increased community financial resources derived from forest-protection activities, that simultaneously lead to substantive improvements in community well-being, promote improved governance (among community institutions, and the communities that hold them accountable), and create tangible incentives for conservation. During this monitoring period, the project has distributed over \$10,168,000 USD to local community revenue-share from REDD+ forest protection activities (See Appendix F).
- Employment opportunities for members of local communities. To date, 108 people from local communities have been hired through the LCFP, among which about a 26% are women.
- Improved natural resource management leading to healthy local ecosystems.

There have been a total of 28,631 community members directly participating in local livelihood activities supported by the LCFP, including the targeted inclusion of women, and youths (see 4.3.1 for details).

Investment in the development of infrastructure and social services using funds generated through the LCFP has wide-reaching positive impacts on local communities, specifically to women, youths, and members of other vulnerable groups (elderly and disabled people).

In 2016, BCP commissioned an Impact Evaluation of the LCFP (Appendix L), aimed at providing baseline estimates of key well-being indicators to serve as benchmarks. The study revealed that the most important assets for community development, according to community members, are (1) health clinics, (2) clean water and (3) tarred roads, while forest resources are not prioritised. As detailed in section 4.3.1, the LCFP invested in the construction of 136 boreholes and wells and supported the development of health centres. Due to the implementation of these resources, among other things, the net impact of the LCFP is positive.

The 2022 follow-up Impact Evaluation (see report in Appendix E) and household mapping survey of the LCFP showed:

- change in household income = 220 % increase compared to the 2016 baseline
- most significant source of income = Agricultural production
- Number of participant and involvement of local community members in REDD+ activities = 222,894 (114,757 Females, Males 108,137) community members have been self-identified as project beneficiaries.
- % community members admitting benefiting from LCFP REDD+ livelihood projects = 75% of households in 2022 compared to 42% in 2020
- % women with a revenue-generating occupation: 55% of women have increased their income level out of livelihood related projects.

Additionally, the LCFP creates positive impacts to local community stakeholders in areas defined by high levels of poverty, and dependence on unsustainable consumption of natural resources. The project is designed to maintain fundamental ecosystem services (provisioning, regulating, supporting) for the communities living in and far beyond the project area.

For these reasons, the project is achieving short and long-term positive impacts for local community stakeholders. These impacts are demonstrable and documented through a variety of sources, including feedback, and quantitative and qualitative data.

4.1.4 Protection of High Conservation Values (CM2.4)

The project area contains various sites of primary significance for the community well-being by providing ecosystem services in critical situations (HCV type 4), basic needs (HCV 5), and traditional and cultural heritage (HCV 6)³¹. (see HCV report in Appendix L for full details). There are multiple springs, streams and rivers within the project areas that are of primary importance for the communities either by providing drinking water and fishing ground and as such these are considered as HCV 5. The riverine forests and forests on slopes of the Luangwa catchment are therefore essential to maintain the provisioning services of these water courses, through filtering and water purification and the regulation of erosion, flood and sedimentation, therefore fulfilling the criteria of HCV 4 in addition to being the areas required to manage the HCV 5 stated above (HCV 5 Management Areas). The REDD+ activities implemented to tackle the drivers of deforestation and to provide climate change adaptation to communities and biodiversity are therefore perfectly suited to maintain the HCV 4 and 5 present in the LCFP.

The LCFP also contains multiple shrines and burial sites that are considered as HCV 6. The preservation of intact habitats in the surroundings of these sites of critical cultural and traditional importance for the local communities are equally suitable as HCV 6 management measures. The agreements signed with the communities setting up the REDD+ activities implementation and co-management, are ensuring that the local communities can freely access these sites.

4.2 Other Stakeholder Impacts

4.2.1 Mitigation of Negative Impacts on Other Stakeholders (CM3.2)

Other Stakeholders, as identified during the stakeholder analysis and mapping exercise prior to validation, are hunting concession holders that operate within the REDD+ project areas and communities from non-project participating chiefdoms adjacent to the REDD+ Zones. All hunting concessionaires have been consulted regarding the project and understand the benefits it will bring. In the same vein, all non-participating chiefdoms with communities living adjacent to the REDD+ Zones have been consulted and engagements and sensitisation in these chiefdoms has continued.

No negative impacts on hunting concession holders is expected from project activities and therefore no mitigation measures have been implemented. Similarly, the well-being of non-project participating chiefdoms adjacent to the REDD+ Zones are not expected to be impacted negatively by the project.

³¹Brown, E., N. Dudley, A. Lindhe, D.R. Muhtaman, C. Stewart, and T. Synnott (eds.). 2013 (October). Common guidance for the identification of High Conservation Values. HCV Resource Network.

4.2.2 Net Impacts on Other Stakeholders (CM3.3)

The project appears to have no negative impacts on the Other Stakeholder groups. On the contrary, the protection of the environment by the REDD+ activities benefit the wildlife (see section 5) which is the basis of the Other Stakeholders' livelihood. Therefore, the well-being of the Other Stakeholders is improved through the LCFP.

4.3 Community Impact Monitoring

4.3.1 Community Monitoring Plan (CM4.1, CM4.2, GL1.4, GL2.2, GL2.3, GL2.5)

Table 16. Details and Results of the Community Monitoring Plan. Vulnerable group encompasses elderly (80+), youth and differently abled people.

Category	Indicators	Community Groups	Methods	Frequency	Results
Demographic Data (Household Census Survey & Mapping)	Total Population	All	Household Registration per Chiefdom	2-5 years	A total population of 222,894 have self-registered in the 17 Chiefdoms. - Females: 114,757 - Males: 108,137
	Women and other Vulnerable Groups (elderly (80)) Total Beneficiaries	Females & Vulnerable Groups	Household Mapping Survey: Total number of females and vulnerable people who self-reported benefits from the projects	2-5 years	A total of 60,302 females and 6,752 vulnerable groups have self-reported to have been receiving benefits from different projects. Hand Pumps: - Females: 13,968 - Vulnerable Groups: 1,247 Clinics: - Females: 9,111 - Vulnerable Groups: 828 Teacher Houses - Females: 1,843 Schools: - Females: 17,191 - Vulnerable Groups: 1,280 Goats: - Females: 849 - Vulnerable Groups: 143 Farming Inputs:

Category	Indicators	Community Groups	Methods	Frequency	Results
					- Females: 6,412 - Vulnerable Groups: 437 Boats: - Females: 1,999 - Vulnerable Groups: 157 Maize Shellers: - Females: 2,622 - Vulnerable Groups: 227 Hammer Mills: - Females: 4,697 - Vulnerable Groups: 371 Police Posts: - Females: 271 - Vulnerable Groups: 47 Iron Sheets: - Females: 1,339 - Vulnerable Groups: 196
Socio-Economic	Household Income	All	Social Monitoring Survey: qualitative and quantitative impact assessment from a representative sample of community members across the project landscape compared against the baseline.	Every 2 years	The 2022 data showed that average household income increased 220% over the five-year period with agricultural production, alternative sources and forest-based products being the key sources of income for the households (Appendix E).
	Household Wealth Indicators	All	Social Monitoring Survey	Every 2 years	Most commonly owned assets among the households are hoes (97%), axes

Category	Indicators	Community Groups	Methods	Frequency	Results
					(92%), cell phones (88%) and residential buildings (84%) (Appendix E)
	Food and Nutrition Security	All	Social Monitoring Survey	Every 2 years	53.4% of the sampled households had experienced periods of hunger during the sample year (July 2021-June 2022), mostly in November-February when most households have depleted their food reserves from the previous farming season (Appendix E)
	Project success rating – are community members benefiting from the project?	All	Social Monitoring Survey	Every 2 years	Self-reported beneficiaries from LCFP in the initial 12 partner chiefdoms increased from 42% in 2020 to 75% in 2022, with a higher proportion of Male Households (80%) than Female Households (69%). On the other hand, considering the short period that the LCFP had been implemented in the 5 new Chiefdoms, only 8% of the households reported having benefited from the REDD+ projects (Appendix E)
	Community Perception towards Forest Conservation and Climate Change	All	Social Monitoring Survey	Every 2 years	80% of households approve the implementation of LCFP's activities, compared to 72% in 2020 and 67% in 2016 85% would support the expansion of community forest under REDD+ management compared to 68% in 2020 (Appendix E)

Category	Indicators	Community Groups	Methods	Frequency	Results
Project Specific Impact Assessments	WASH Projects	All Community Groups	Household Mapping Survey	2 - 5 Years	26,915 self-reported beneficiaries • Total Households (HHs): 5,745 • Males: 12,947 • Females: 13,968 • Vulnerable Groups: 1,247
	Health Projects	All Community Groups	Household Mapping Survey	2 - 5 Years	17,712 self-reported beneficiaries from Clinics • Total HHs: 3,922 • Males: 8,601 • Females: 9,111 • Vulnerable Groups: 828
	Education Projects	All Community Groups	Household Mapping Survey	2 - 5 Years	3,662 self-reported beneficiaries from Teacher Houses: • Total HHs: 712 • Males: 1,819 • Females: 1,843 36,244 self-reported beneficiaries from Schools: • Total HHs: 6,647 • Males: 19,053 • Females: 17,191 • Vulnerable: 1,280
	Community Livelihood Projects	All Community Groups	Household Mapping Survey	2 - 5 Years	28,631 self-reported beneficiaries: Maize Shellers: • Total HHs: 1,281 • Males: 2,482 • Females: 2,622 • Vulnerable groups: 227 Hammer Mills:

Category	Indicators	Community Groups	Methods	Frequency	Results
					• Total HHs: 1,917 • Males: 4,506 • Females: 4,697 • Vulnerable Groups: 371 Goats: • Total HHs: 294 • Males: 813 • Females: 849 • Vulnerable Groups: 143 Farming inputs: • Total HHs: 2,707 • Males: 6,250 • Females: 6,412 • Vulnerable Groups: 437
Climate Smart Agriculture – Livelihoods	Trainings	Smallholders Disaggregated by Training Type	Tracking and Collection of Training Reports and Attendance Lists of Approved Trainings	Annually	Climate Smart Agriculture (LFs): • Total: 1386 • Male: 883 • Female: 503 Horticulture: • Total: 172 • Male: 121 • Female: 51 Beekeeping: • Total: 225 • Male: 91 • Female: 134
	Hectares (ha) under Climate	Small holders	Annual Lead Farmers Survey– Field Based Data (ODK)	Annually	408.5 ha converted to Climate Smart Agriculture / Conservation Farming.

Category	Indicators	Community Groups	Methods	Frequency	Results
	Smart Agriculture				
	Yield Assessment Survey	Small Holders	Annual survey comparing yields under Climate Smart Agriculture (CSA) and Conventional Agriculture by District	Annually	Mean maize grain yields increased from 1.8 tons/ha under conventional agriculture during the baseline year to 4.9 tons/ha and 4.0 tons/ha under CSA in 2020/2021 and 2021/2022 farming seasons respectively (Appendix H)
	Honey Data	Honey Farmers	Field based data collection by Nature's Nectar, verified by BCP and Communities	Annually	• Total Beehives: 9622 • Total Kgs Harvested: 6,020 kgs • Total Payout: 7,900 USD
Finance and Governance	Direct Local Employment	All	Job opportunities are advertised in the Communities through the CFMGs, Headmen and Chiefs. Direct Employment of Community Members – tracked through HR Database and Payroll	Ongoing	108 Community Members have been hired from LCFP Chiefdoms. Chiefdom Hires: 30 • Males: 25 • Females: 5 Community Scouts: • Total: 78 • Males: 55 • Females: 23
	Funds Disbursements as per Benefit Sharing Mechanism	CFMG's	VCU's sold and funds distributed to CFMG's as established by the Benefit Sharing Mechanism	Semi-Annually	\$10,168,000 distributed to local communities as community revenue-share and over \$1,971,122 spent in community project support
	Community Engagement	All Community Members	Tracking of Trip Reports and Attendance Lists from all	Ongoing	In 2021 alone more than 250 meetings where held comprising over 10,673

Category	Indicators	Community Groups	Methods	Frequency	Results
			Village, VAG and CFMG Meetings		participants (5,665 females and 5,008 males)
	Annual Work Planning and Needs Assessment	All community members	Annual Work Plans and Needs Assessments are completed and signed off by Zone Cooperatives and GRZ Partners, ensuring Communities are included in the Governance and Implementation of the Project.	Annually	Needs Assessments Reports and Annual Work Plans are developed at CFMG level and completed yearly for sign-off by all relevant authorities.
HCV's	Monitoring of High Conservation Values (HCV's)	All Chiefdoms	High Conservation Values Survey and Assessment	Ongoing	Consultations with the Community Scouts and Community Members have confirmed the presence of HCV 1 - 6 attributes within the Chiefdoms.

4.3.2 Monitoring Plan Dissemination (CM4.3)

LCPD distributes the results of its monitoring plan through three primary distribution mechanisms. First, is through BCP's communications and marketing strategy. This strategy is oriented towards sharing important impacts and success stories at a local, national and international level through BCP's websites, as well as newsletters and social media posts. It also includes more locally oriented communication channels such as hosting weekly radio shows. The second mechanism is the distribution of the results of key surveys and impact reports (See Appendix J). Specifically, the results of the social monitoring survey (the Income and Perception Survey), the household census and the yield assessment survey are shared with the CFMG's and VAG's. To make the information more accessible to communities, executive summaries are translated into local languages and full documentation made available at CFMG offices. Finally, as part of adaptive management approach and BCP's regular information sharing and consultation strategy, impacts are discussed and shared with stakeholders (such as GRZ partners), CFMG's and communities during quarterly planning meetings. Building on these quarterly planning meetings, BCP and the CFMG's also implement an annual needs assessment and annual working planning process in quarter 4 of every year. As part of this process, BCP, the CFMG and representatives from other stakeholders visit all of the VAGs in each chiefdom to share progress on community projects and solicit feedback and inputs on projects for the following year. Through this participatory planning process, projects are evaluated and selected for the upcoming year and progress checks with communities planned for to ensure a transparent implementation process.

4.4 Optional Criterion: Exceptional Community Benefits

4.4.1 Short-term and Long-term Community Benefits (GL2.2)

The project has delivered short-term and long-term human well-being benefits across all three areas of well-being outlined in the Attacking Poverty framework³². These benefits include:

Opportunities: The project has created material opportunities for wealth creation and well-being through the payment of conservation fees, creation of jobs, alternative livelihoods through beekeeping activities, conservation farming activities and investment in infrastructure projects and improved educational conditions.

During the monitoring period, the following benefits were delivered:

- More than \$10,000,000 was disbursed in conservation fees during this monitoring period, in addition to the initial investment during the project development phase (under USAID funds):
 - Employment of 78 community scouts
 - Health services were improved for 17,712 people
 - Educational conditions were improved for 36,244 people
 - Improved access to water delivered to 26,915 people
- There were 1089 participants in the honey production program and income diversification was achieved through beekeeping activities

³² Lawlor, K., Madeira, E.M., Blockhus, J. and Ganz, D.J., 2013, Community Participation and Benefits in REDD+: A Review of Initial Outcomes and Lessons, *Forests*, 4(2), 296-318.

Security: The project has enhanced the communities' security including tenure security, livelihood security, and capacity to adapt to climate change.

- BCP and its Government partners have supported the first CFMGs in the country to apply for the form 7 as a step towards obtaining forest user rights, including carbon. During the monitoring period, 5 of the 17 CFMG's have applied for form 7 and obtained forest user rights.
- The project has implemented project-based activities to increase the resilience of local communities through improved farming practices. There were 396 participants in the climate smart agriculture program.

Empowerment: The project has facilitated the empowerment of individuals and communities to participate in decisions related to use of conservation funds. A yearly needs assessment using PLA (Participatory Learning Approach) methods has been developed with each of the VAGs in order to identify community projects and livelihood activities for the upcoming year with direct input from community members/VAGS. Using a problem tree analysis community projects have been selected, budgeted and planned in the CFMG's AWP's. These projects were then communicated back to the VAGs and communities after selection.

Overall, as per the 2022 follow-up Impact Evaluation of the LCFP, 75% of households sampled acknowledged benefitting from LCFP REDD+ activities, therefore demonstrating that the LCFP has a positive impact for the well-being of a large proportion of the community members.

Expansion of all these well-being benefits, including expansion of beekeeping, conservation farming activities and direct cash dividends are expected to continue in the future for the entire life of the project.

Table 17. Description of expected short-term and long-term impacts of project activities

Activity	Short-Term Impacts	Long-Term Impacts	Net Impacts
Climate Smart Agriculture & Agro-Forestry	• Higher yields • Lower costs • Early land preparation helps to spread the labour which is freed to other economic and social ventures. • Higher incomes • Food security • Crop diversification • Financial diversification • Technical empowerment, / skills and / knowledge • Awareness of environmental issues	• Reduced clearing due to higher productivity and/ efficiency • Resilient climate change and economic adverse communities in the face of climate change and economic adversity • Conservation incentives • Communities keep the land and soil healthy for longer, hence no abandoning of land in search of new fields. • Development of skills base and local knowledge in terms of conservation farming and natural resources management • Market growth around sustainable agriculture	Positive

Social Services / Infrastructure	• Direct benefits on human well-being – provision of improved access to clean/safe drinking water, improved school and/or healthcare facilities etc. • School infrastructure support programs improve local educational services.	• Increase in Healthier community health with higher indicators for human well-being • Improved governance of local leaders/institutions • Empowered community's / community accountability • Reduced incentives to clear land for new to establish new communities – communities “build out” from existing areas of development, around existing social services. • Tangible incentives for conservation – promotes REDD+ as PES model with demonstrable impacts on communities and conservation	Positive
Community outreach	Increased awareness / knowledge of environmental issues	Future generations of environmentally-mindful citizens	Positive
Apiculture	• Increased honey production • Improved incomes • Financial diversification • Technical empowerment, business skills and knowledge	• Reduced deforestation • Enhanced carbon sinks • Enhanced wildlife habitats • Sustained livelihoods	Positive
Horticulture	• Increased revenue stream and income • Improved nutrition at household level	• Improved Livelihoods at household level • Reduced risks of deforestation as other alternative livelihoods are able to support the needed incomes.	Positive
Poultry	• Contribute to household food and nutrition security • Contribution to gender equity • Improved revenue stream • Technical empowerment; business skills and knowledge in poultry management	• Sustainable livelihoods • Reduced risks on deforestation and wildlife habitats • Resilient and adaptable communities against impacts of Climate Change	Positive

4.4.2 Marginalized and/or Vulnerable Community Groups (GL2.4)

Table 18. Net positive impact on marginalised and vulnerable community groups

Community Group	Woman and/or elderly- headed households
Net positive impacts	As per income perception survey (Social Monitoring Survey 2022; Appendix E), 69% of female-headed households sampled and 11% of youth-elderly-physically impaired people reported to benefit from the LCFP livelihood projects. As described in section 4.3.1, the investment in infrastructure development results in improved access to critical assets (water, health, education) for all community members including women and vulnerable groups respectively. The project results in ecosystem services maintenance which is a critical indicator of well-being for each and every community group.
Benefit access	All community members contribute equally in the Need Assessment process held at the chiefdom level to decide on revenue expenditure (see section 4.4.4: benefit sharing mechanism). BCP has also worked with CFMG's to develop bylaws concerning the use of funds from REDD, these bylaws require CFMGs to work within the approved budgets and accurately account for expenditures related to all approved projects. BCP is also working hard to reduce any incentives that may cause destructive or corrupt activities. These measures seek to address the risk of elite capture and that women and elderly headed households are excluded from decision-making and receipt of benefits.
Negative impacts	The project is specifically designed to provide benefit to these members by ensuring their active participation in project selection, and by providing livelihood opportunities. No negative impacts were anticipated. However, the project continues to monitor impacts to these members.

4.4.3 Net Impacts on Women (GL2.5)

The LCFP recognizes the importance of mainstreaming gender in the promotion of sustainable use and management of natural resources under a REDD+ project. The project ensures that men and women are equally able to access, participate, and benefit from natural resource management projects, and that gender inequalities are reduced or eliminated.

BCP implements activities to challenge gender stereotypes, support women to take leadership roles in natural resources, Economic opportunities for women are encouraged, through livelihood empowerment support and linkages to stable markets.

The project has benefited women in the following ways: increased income and assets, increased knowledge and skills among women, improvement in the overall health of women and their families within the Project Zone, and reflection of women's needs/wants/desires in the need assessment process held at the chiefdom level to decide on revenue expenditure (for detailed information refer to section 4.3.1).

During the monitoring period, it is estimated that improved outcomes were achieved for 9,111, 17,191 and 13,968 women, in the areas of health, education and improved water respectively. As stated earlier, 69% of female-headed households self-reported as beneficiaries of LCFP livelihood projects. Other benefits that accrued to women include:

- Paid labour and direct employment: Employment of 23 female community scouts
- 58 women participating in the climate smart agriculture program
- 688 female community members received training which improved their skills/knowledge

Number of women for whom well-being was improved as a result of project activities during the monitoring period was 60,302.

4.4.4 Benefit Sharing Mechanisms (GL2.6)

The project's benefit sharing model is based on the principle that benefits from community-based natural resource management (CBNRM) utilised on communal projects have more advantages over systems where revenue is spent on individual community members. Using CBNRM as a tool for local area development gives community members flexibility and decision-making power over the use of their natural resources. This approach supports the building of strong local governance institutions that facilitate the delivery of communal benefits and encourage wide participation in decision-making.

BCP is aware that the potential revenue from REDD+ could be substantial and in some instances exceed the revenue from hunting concessions that the communities are accustomed to managing. In this regard, BCP has worked with participating communities and the Department of National Parks and Wildlife to strengthen financial systems for managing community revenue from natural resource management.

Community consultations to review and revise these guidelines follow a bottom-up system, starting at the lowest governance structure; the Village Action Group (VAG), up to the Community Forest Management Groups (CFMG). Most importantly, the benefit sharing guidelines recommend that communal projects selected by the VAGs are presented to the CFMG where they are consolidated and the final decision on communal projects to implement from revenue generated from REDD is made.

The basic steps of the process are as follows:

1. **Community General Meeting:** Community meetings for all community members over the age of 18 are convened by BCP and the CFMG where a cheque for the revenue from REDD is presented.
2. **VAG Meetings:** The VAG committee convenes a meeting with all members of the community over the age of 18 to review community needs and compile a list of communal projects to implement. Budgets are developed at VAG level and these, along with records of meetings held, are submitted to the CFMG.
3. **CFMG Meeting:** A board meeting for the CFMG is convened at which representatives from the individual VAGs present to the board their list of selected projects and budgets. Using PLA tools, the board considers each selected project on its merits and perceived benefits to the community. The board

makes a final decision on which communal projects to implement and the final decision is communicated to the VAGs.

BCP has also worked with CFMG's to develop bylaws concerning the use of funds from REDD, these bylaws require CFMGs to work within the approved budgets and accurately account for expenditures related to all approved projects. This serves as a safeguard against elite capture. With the successful implementation of some community projects, community members are showing more confidence in the system to deliver development in their villages.

4.4.5 Governance and Implementation Structures (GL2.8)

The LCFP is implemented through CFMGs as local governance institutions with authority operating at a Chiefdom level. CFMG's are composed of elected representatives from each VAG within a chiefdom. VAGs are sub-Chiefdom level governance structures and are composed of groups of villages. VAG and CFMG representatives are locally-elected positions. The full and effective participation of community members in decision-making and implementation, including decisions related to benefit sharing, is facilitated through these self-governance structures.

LCFP implementation planning and decision making process starts at VAG level where all community members of 18 years and above are invited to participate. Whilst key decisions relating to LCFP are made by the CFMG, extensive consultation is achieved at village level through the VAG. Decisions made at the CFMG level with consultation from the VAGs and traditional leaders include the approval of the project zones and community forest reserve boundaries, management, by-laws and REDD+ revenue distribution.

These local governance structures, including the CFMG are aligned with the Wildlife Act No. 14 of 2015 and have been recognized as CFMG as per the Forest No 4 of 2015 of the Laws of Zambia. BCP has signed MoUs, and agreements with each participating community through the CFMG, which obliges BCP to generate and trade in carbon offsets on behalf of the community for 30 years.

4.4.6 Smallholders/Community Members Capacity Development (GL2.9)

BCP has invested in developing the capacity of community members and local self-governance institutions through training, Needs Assessment and Annual Work Planning processes as thoroughly described in the respective sections above and Appendices A and J.

Table 19 summary of capacity building conducted for community members and stakeholders between 2019 - 2022

Training	Male	Female	Total
Agriculture	883	503	1386
Horticulture	121	51	172
Beekeeping	91	134	225
Conservation	374	128	502
Total	1469	816	2285

5 BIODIVERSITY

5.1 Net Positive Biodiversity Impacts

5.1.1 Biodiversity Changes (B2.1)

Table 20. Biodiversity Changes: Habitat connectivity/Wildlife Corridors

Change in Biodiversity	Habitat connectivity/Wildlife Corridors
Monitored Change	During the monitoring period, the project has maintained a vital section of intact forest along the Luangwa river to the Lower Zambezi ecosystem, connecting five National Parks. A scarce number of encroachments took place within the LCFP REDD+ area, as demonstrated by aerial patrols, remote sensing (Figure 3) and supported by leakage plot monitoring (Appendix B).
Justification of Change	The project area is in excess of 1 million ha and maintains critically threatened forest habitats in some of Africa's last remaining wilderness areas, the Luangwa valley and the Lower Zambezi systems. However, this area is progressively threatened by a growing human population and subsequent deforestation for subsistence farming and charcoal production. The implementation of REDD+ activities is enhancing the protection of these areas while providing livelihood alternatives that have resulted in the successful avoidance of important wildlife habitat loss, as demonstrated through remote sensing and encroachment and leakage monitoring. The documentation by the Zambian Carnivore Programme of long distance dispersal events of African wild dogs (<i>Lycaon pictus</i>) across the LCFP during this monitoring period (Figure 3) is an illustration of the relevance to maintain this connectivity and ecological functionality of this protected area network through improved management³³. Likewise, the movement data of the three white-backed vultures (<i>Gyps africanus</i>) tagged in LCFP project area by Birdwatch Zambia and BCP in August 2022 demonstrates the importance of LCFP in the maintenance of a critical wildlife corridor (Figure 3).

³³ <https://bcp.earth/news-and-press/wild-dogs-are-crossing-borders-using-zambias-protected-wildlife-corridors/>

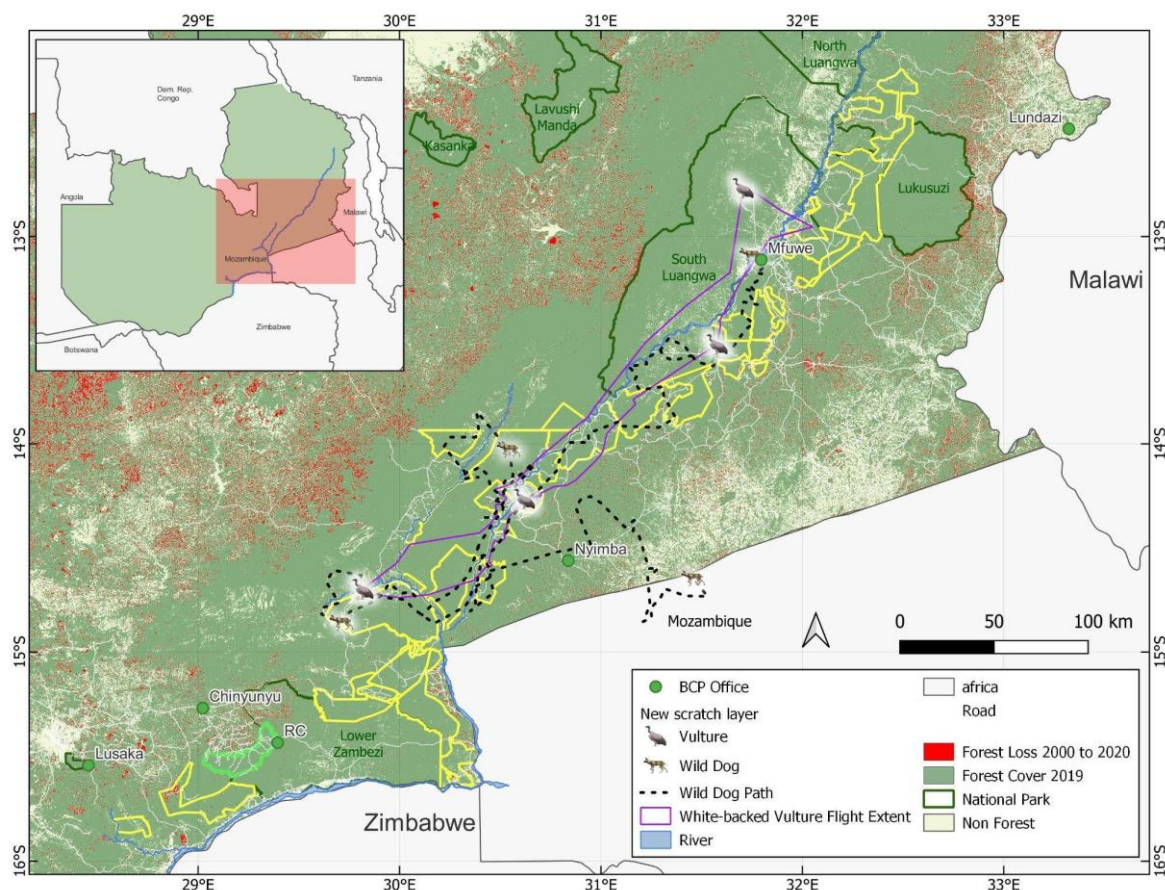


Figure 3. Habitat connectivity and wildlife corridor maintenance within the LCFP landscape

Table 21. Biodiversity Changes: Poaching pressure

Change in Biodiversity	Poaching pressure
Monitored Change	The project is contributing to the reduction of poaching pressure in the project area, especially in areas where no conservation organisations or private actors operate. This, in turn, would result in healthier wildlife populations, with populations recovering or remaining stable. The project activities, namely the BCP-support to surveillance and law enforcement patrols conducted by scouts in community forests and private ranches, resulted in the observation of numerous poaching incidents (snare, firearms, carcasses, encounter with poachers, gun shots, etc). In 2021 for instance, a total of 90 snares were removed, 2 firearms were confiscated and 8 poachers were successfully apprehended by the BCP-supported scouts. (Appendix G).
Justification of Change	Using the 'without-project' scenario, surveillance and law enforcement patrol efforts would have been significantly reduced, in parts of the LCFP under community management where no conservation organisations

	operate (eg. Conservation South Luangwa) and that depend almost exclusively on the Department of National Parks and Wildlife (DNPW) and CFMG resources. The use of Spatial Monitoring And Reporting Tool (SMART) was employed in 2020 for the planning and management of patrols which greatly improved the efficiency. SMART is not currently being used systematically and at the scale of the LCFP by non-BCP-supported community scouts, or by DNPW Wildlife Police Officers. BCP's use of performance-based incentives and payments for forest conservation also serves to discourage poaching by community members.
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Table 22. Biodiversity Changes: Wildlife populations

Change in Biodiversity	Wildlife populations
Monitored Change	At the start of the project some areas had depleted wildlife populations due to heavy poaching, resulting in 'empty forests'. In these areas, it is expected that the abundance of fast reproducing species (r-organisms) that are sensitive to poaching pressure will increase in the short-term. Similar trends will be observed for slower reproducing species (k-organisms) in the medium- to long-term. It is also expected that some species will increase their distribution range, by occupying areas that become more suitable following the reduction of poaching pressure. Habitat suitability may also increase along with the secondary succession observed in degraded lands. Additionally, forest-dwelling species are experiencing a reduced rate of habitat loss and fragmentation as a result of REDD+ activities. Forest loss in the project area, measured against the without-project scenario, would have resulted in a total of 34,509 ha's lost between the period 2015-2021. Nevertheless, across the project area, only 3097 ha were actually lost during this same period. As an example, the Mopane specialist Lilian's lovebird (<i>Agapornis lilianae</i>; Near Threatened), whose range is centred around the Luangwa and Zambezi basins, is highly reliant on old, tall and large mopane trees to breed³⁴. Without appropriate management practices, suitable breeding sites would become increasingly scarce through the clearance of Mopane stands for illegal logging, charcoal production and farming.
Justification of Change	Due to a combination of improved land management, surveillance and law enforcement patrols, wildlife populations are expected to have increased in some areas. Such increases in abundance have been observed in Munyamadzi with transect surveys conducted annually since 2016 (see Biodiversity Monitoring Annual Reports in Appendix D)

³⁴ MZUMARA, T.I., Martin, R.O., Tripathi, H., Phiri, C. and Amar, A., 2019. Distribution of a habitat specialist: Mopane woodland structure determines occurrence of Near Threatened Lilian's Lovebird *Agapornis lilianae*. *Bird Conservation International*, 29(3), pp.413-422.

Table 23. Biodiversity Changes: Threatened species population

Change in Biodiversity	Threatened species populations
Monitored Change	The populations of African wild dogs (Endangered), lions (Vulnerable) and leopards (Vulnerable) remained stable. The population of other threatened species of large mammals such as elephants (Endangered), hippos (Vulnerable), giraffes (Vulnerable), and of birds such as grey crowned cranes (Endangered), vultures (hooded, white-backed, white-headed: Critically Endangered; lappet-faced: Endangered), southern ground-hornbills (Vulnerable) and Tawny eagle (Vulnerable) are expected to have remained stable as well.
Justification of Change	Due to a combination of improved land management, effective surveillance and law enforcement patrols, human-wildlife conflict mitigation and sensitization, implemented by the various organisations active in the Luangwa valley (Conservation South Luangwa, Zambian Carnivore Programme) and supported by BCP, as well as operators on private lands, the population of endangered species is expected to remain stable and will experience an increase in the long-term, especially for the carnivores (African wild dogs, lions, leopards) whose population will take advantage of a higher prey abundance in a less human-impacted (through snaring) area. Likewise, vulture populations will benefit from a healthier wildlife population (i.e. food availability) as a result of project activities and the anti-poisoning program started in 2022 in collaboration with BirdWatch Zambia (see MOU in Appendix D). Due to the significant law enforcement efforts supported by Conservation South Luangwa (CSL), and other organisations including BCP, a reduction of snaring pressure has been reported in recent years, with no lion being snared in 2020, no giraffe and wild dogs snared in 2021, and a reduction in elephant poaching. In 2018 more than 60 elephants were killed within the CSL jurisdiction whereas in 2020 less than 20 were killed.³⁵³⁶ LCFP is improving the protection of approximately 326km of the Luangwa River stretch, therefore contributing to the protection of the largest population of hippos globally³⁷. The protection of the forest resulting from the project significantly contributes to the conservation of the southern ground-hornbill as this

³⁵ Conservation South Luangwa. Annual Report 2020. <https://cslzambia.org/newsletters-annual-report>

³⁶ Conservation South Luangwa. Annual Report 2021

³⁷ Lewison, R. & Pluháček, J. 2017. *Hippopotamus amphibius*. *The IUCN Red List of Threatened Species* 2017. <https://www.iucnredlist.org/species/10103/18567364#population>

	species is primarily limited by the availability of old trees with natural cavities for nesting.³⁸ The extensive monitoring programme using camera traps and spatially explicit capture-recapture models carried out by the Zambian Carnivore Programme (ZCP) shows a stable population of carnivores in the Luangwa valley.³⁹
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5.1.2 Mitigation Actions (B2.3)

Mitigation measures were put into action for livelihood restoration, community support, surveillance and law enforcement patrols. According to the validated PD document, there were no anticipated negative impacts of the project activities on biodiversity and therefore no specific mitigation measures were taken. The most common negative impacts a REDD+ project could have on biodiversity include (i) the displacement of human activities to forests of higher biodiversity value, or (ii) to non-forest areas of high biodiversity value, and (iii) a failure to protect the wildlife due to the emphasis on deforestation⁴⁰.

In the case of LCFP, it is unlikely that a 'biodiversity leakage' will occur. This is because areas of high conservation value are in the National Parks which benefit from active surveillance and protection from governmental (DNPW) and non-governmental organisations (Conservation South Luangwa, Conservation Lower Zambezi, Frankfurt Zoological Society, Wildlife Crime Prevention Zambia). Furthermore, conservation organisations and private operators work to manage the natural resources in the GMAs and open lands (eg. COMACO, Nsefu Wildlife Conservation Foundation, Game Rangers International, Zambezi Wildlife Trusts, Luambe Camp and Conservation, Makasa Safaris, Munyamadzi Game Reserve).

BCP massively invests in surveillance and law enforcement activities, significantly improving the effective management of the Game Management Areas by the DNPW and CFMG's, especially in areas that were not already benefiting from other stakeholders' support. For instance, BCP is supporting a roadblock for wildlife and forest products on the main road linking the South Luangwa National Park and the centre city of Eastern Province, Chipata, the fifth largest city of the country. This is serving as a deterrent for illegal hunting on a scale surpassing the Project Area (although not surmountable). In addition, the protection of wildlife is further secured through the signing of Community Covenants and similar agreements that oblige both BCP and participants to comply with project guidelines and connect community benefits to forest protection efforts.

Despite no negative impacts relating to biodiversity within the project zone, an HCV 1-3 assessment was conducted in 2022 in all chiefdoms. The result of the assessment is presented in Appendix L. The survey confirmed that the project zone contains multiple HCV 1-3 attributes and that the conservation and community support activities implemented by BCP are adequately managing them.

The project activities are designed to reduce the human pressure on the environment, therefore contributing to the restoration of natural processes as the main driver shaping the ecosystem composition, structure and dynamics. In order to achieve this, and ensure the support of the communities who depend on the forest to

³⁸ Mabula Ground Hornbill Project. <https://ground-hornbill.org.za>

³⁹ Zambian Conservation Program. <https://www.zambiacarnivores.org/annual-reports>

⁴⁰ Narasimhan, P., Starr, I., Hayward, J., Noponen, M. and Durbin, J. (2014). *Guidance for the use of the CCB Standards*. Washington, DC: Climate, Community and Biodiversity Alliance and the Rainforest Alliance.

meet certain basic needs, BCP invests in sustainable agriculture, alternative livelihoods and development projects. As such, the LCFP is expected to simultaneously deliver social and biodiversity benefits, together with GHG emissions reductions.

5.1.3 Net Positive Biodiversity Impacts (B2.2, GL1.4)

REDD+ projects intend to preserve forested land from the deforestation activities observed in its direct vicinity. Therefore, the project activities are reducing the loss of suitable habitats for forest-species and other organisms relying on them. The without-project scenario would result in habitat contraction, fragmentation and ultimately, biodiversity loss. As there is no biodiversity leakage foreseen and observed for the LCFP, the with-project scenario is therefore providing a net biodiversity benefit.

Southern Ground-hornbills and Lilian lovebirds are examples illustrating the net positive biodiversity benefit under the with-project scenario. Both species rely on old, large and tall trees for nesting and breeding. Without the project, such trees would be targeted for charcoal production or simply disappear through land clearance for agriculture expansion.

The southern ground-hornbill is listed as Vulnerable on the IUCN Red List of Species and one of the main limiting factors to its distribution and survival is the availability of suitable trees to breed⁴¹.

The Lilian's lovebird, is listed as Near-Threatened and is endemic to the Zambezi basin and tributaries. It is widely referred to as a mopane specialist, breeding preferentially in large, tall Mopane trees which is a dominant species in the LCFP landscape⁴². The avoided deforestation resulting from the project is securing the maintenance of suitable corridor habitats connecting five National Parks centred around the Luangwa River and its tributaries, while the conservation and community support activities are improving its ecological functionality.

In March of 2022, the Zambian Carnivore Programme posted on its social networks the incredibly long journey of over 1200 km several female wild dogs took. Equipped with tracking devices, the dispersing group started the journey from South Luangwa National Park down to the vicinity of Lower Zambezi National Park and across the Mozambique border (Figure 3) and then finally back to Zambia into the Lower Zambezi National Park (Matt Becker, pers. Comm.). During this long distance dispersal, the wild dogs mainly travelled through LCFP REDD+ zones. This stresses the importance of improving the management of these areas that are crucial for maintaining the ecological function of this protected area network which is part of the Lower Zambezi-Mana Pool and ZIMOZA Transfrontier Conservation Areas (TFCA). Therefore, the LCFP is assisting biodiversity adapt to the predicted impact of climate change by securing access to critical water sources within and between River catchments (Luangwa and Zambezi).

⁴¹ Kemp A.C. & Begg K.S. 1996. Nest sites of the southern ground hornbill, *Bucorvus leadbeateri*, in the Kruger National Park, South Africa, and conservation implications. *Ostrich* 67: 9-14.

⁴² MZUMARA, T.I., Martin, R.O., Tripathi, H., Phiri, C. and Amar, A., 2019. Distribution of a habitat specialist: Mopane woodland structure determines occurrence of Near Threatened Lilian's Lovebird *Agapornis lilianae*. *Bird Conservation International*, 29(3), pp.413-422.

5.1.4 High Conservation Values Protected (B2.4)

The project is reasonably expected not to have negative impacts on biodiversity within the project area and outside as no biodiversity leakage is anticipated. Therefore, none of the High Conservation Value attributes (HCVs) present in the project zone is negatively affected by the project activities. However, HCVs are getting a higher protection level from the project activities.

As explained in section 5.1.2, an HCV assessment was conducted in 2022 in all chiefdoms and showed that the conservation and community support activities implemented by BCP are adequately managing the HCV attributes (see the HCV assessment report in Appendix L).

5.1.5 Invasive Species (B2.5)

As described in the PD, the project does not use invasive species (see below the list of non-native species used in the project). The invasiveness potential of all species used by the project for livelihood activities and agro-forestry have been checked on the Global Invasive Species Database.⁴³

5.1.6 Impacts of Non-native Species (B2.6)

Table 24. *Zea mays* (Maize)

Species	<i>Zea mays</i> (Maize)
Justification of Use	Sorghum spp. is a potential alternative and has been replaced in favour of maize as it produces bigger yields and is not prone to bird predation.
Adverse Effect	No negative evidence documented. It is used extensively in Zambia

Table 25. *Cajanus cajan* (Pigeon pea)

Species	<i>Cajanus cajan</i> (Pigeon pea)
Justification of Use	No known local alternative within native species. It is a nutritious legume.
Adverse Effect	No negative evidence documented. It is used extensively in Zambia.

Table 26. *Mangifera indica* (Mango)

Species	<i>Mangifera indica</i> (Mango)
Justification of Use	<i>Parinari curatellifolia</i> is a potential alternative and is locally utilised but does not have the same production and fruit size and is slow growing.
Adverse Effect	No negative evidence documented. It is used extensively in Zambia

Table 27. *Carica papaya* (Papaya/ Pawpaw)

Species	<i>Carica papaya</i> (Papaya/ Pawpaw)
Justification of Use	No known local alternative within native species with fruit that size.
Adverse Effect	No negative evidence documented. It is used extensively in Zambia

⁴³ Global Invasive Species Database (2021). Downloaded from <http://www.iucngisd.org/gisd/search.php> on 09-12-2021.

Table 28. *Moringa oleifera* (Moringa)

Species	<i>Moringa oleifera</i> (Moringa)
Justification of Use	There are possibly alternative local species that have the same nutritional values, but <i>Moringa oleifera</i> is particularly fast growing.
Adverse Effect	No negative evidence documented. It is used extensively in Zambia

Table 29. *Faidherbia Albida* (Formerly Acacia Albida)

Species	<i>Faidherbia Albida</i> (Formerly Acacia Albida)
Justification of Use	This species is indigenous in Zambia although it may not naturally occur in all project areas. It is valuable for its nitrogen fixing and soil stabilizing functions and therefore contributes to soil fertility improvement. <i>Faidherbia Albida</i> can be used for fodder.
Adverse Effect	No negative evidence documented, it is used extensively in Zambia.

Table 30. *Gliricidia sepium*

Species	<i>Gliricidia sepium</i>
Justification of Use	This species is a multi-purpose nitrogen fixing tree which is a valuable asset to agroforestry and conservation agriculture while aiding soil stabilization, providing fodder for livestock. It is fast growing, produces a lot of biomass which contributes to soil organic matter thereby improving soil fertility. It tolerates repeated cutting or harvesting. The stems of <i>Gliricidia sepium</i> can be used as firewood thereby reducing pressure on fuel wood extraction from the forests.
Adverse Effect	Whilst the species is considered a weed in Jamaica, this is not the case in any other area, including Zambia, and is not listed as invasive on the Global Invasive Species Database. The trees generally show limited seed production outside their native range due to a lack of pollinators and poor conditions for seed germination meaning this risk is extremely low.

Table 31. *Tephrosia candida* and *Tephrosia vogelii*

Species	<i>Tephrosia candida</i> and <i>Tephrosia vogelii</i>
Justification of Use	These are particularly useful drought-tolerant nitrogen fixing species as the rate at which they release nitrogen closely matches crop uptake and reduces loss of nitrogen. <i>Tephrosia</i> is rich in bioactive phytochemicals which have strong insecticidal activities against a wider range of pests – the leaves of <i>Tephrosia</i> are widely used for pest control.
Adverse Effect	In Zambia, no negative effects have been documented.

5.1.7 GMO Exclusion (B2.7)

The LCFP does not undertake afforestation/reforestation activities to generate GHG emissions reductions or removals and does not use GMOs in Conservation Farming and Climate Smart Agriculture. In addition, The biosafety act of 2010 forbids the sale of GMO seeds within Zambia.

5.1.8 Inputs Justification (B2.8)

Agricultural activities undertaken by the LCFP are only involved in conservation farming where a minimum of inputs is being used. Instead, conservation farming promotes organic methods.

Each Lead Farmer receives 100 kg of D Compound – a basal application fertiliser, and 100 kg of Urea as a top dressing fertiliser, mainly for the maize crop.

Table 32. D compound and Urea Fertilizer Products

Name	<i>D compound and Urea Fertilizer Products</i>
Justification of Use	Both products are widely available and farmers have been employing both products for the better part of 40-50 seasons in Zambia and across the wider region. When used correctly and safely in an efficient tillage practice (discussed in greater detail below) efficacy is significant with minimal side effects of the kind that can be realised from prior generations of pesticides, fungicides and herbicides. While the range of organic fertiliser products has both improved in quality and availability, D Compound and Urea are still price competitive, and when used properly, tend to outperform organic fertiliser products. Like organic products when employed in traditional tillage practices are not without their environmental impact. That said, BCP are exploring, with a local company – Albida Agriculture, are testing two of their bio-fertilizer (maize and soya) products with fifty interested and capable Lead Farmers this coming season (2022/2023 farming season). We are exploring a potential transition from chemical fertilisers to comparable bio-fertilisers. Commercial trials show promise, but any sort of new product roll out into the smallholder sector takes time as on-farm conditions can vary from one farm to the next.
Adverse Effect	Minimal due to the tillage practices employed. Both fertilisers are employed in hand hoe basin and or animal draught power rip lines (tram line) minimal tillage practices. The permanent hand hoe basins and or animal draught power rip lines, unlike overall tillage practices enables a farmer to better place and protect both products from the effects of rain and wind borne erosion effects as well as mitigating third party direct contact with humans, livestock and or wildlife unlike can be the case with traditional overall tillage practices, described in the following paragraph below. Overall tillage practices such as hand hoe ridging and or animal/tractor ploughing is prone to significant water and or wind erosion effects as 100% of the top soil is inverted to establish crops. This inversion invariably leads to product run offs and could increase product contact with humans, livestock, and or wildlife. Run offs as a result of overall tillage practices can also transport nitrogen and phosphorous nutrients (both synthetic and organic) from fields into waterways – leading to eutrophication. It is for these reasons, among others, that BCP promote the conversion of overall tillage practices to the Conservation Minimal Tillage practices. With the minimal tillage practices the projects are exposing and training farmers on the D compound product is placed directly towards or on the bottom of the 15

	cm deep planting basin or rip line. Once placed in the basin or rip line, the product remains in a sedentary state at the bottom of the planting basin or rip line. The product is then immediately backfilled and covered with approximately 3-5 kg of top soil that is placed directly on top and across the product once the application operation is done and prior to the first planting rains and the sowing operation that follows. Seeding is done directly in the basin and or rip, directly over the product so emerging tap roots can quickly access the product before dissipation quickly occurs from rainwater soak into the soil. In the case of Urea, farmers ensure that the product is also covered up with top soil, and is placed adjacent to the knee-high (50-55 cm high) maize crop. Covering the product with top soil coupled with the physiological stage of the maize and or soya crop at the time of product application further mitigates potential negative run offs and direct contact with humans, livestock, and wildlife. Further, covering urea with top soil reduces the Nitrogen Oxidation process which can lead to the production of Nitrous Oxide. This, in turn significantly reduces the emission of Nitrous Oxide from urea application Each season, Lead Farmers, who in turn train follower farmers are required to attend a whole gamut of refresher training, including the correct and safe application of both products into their planting basins and or rip lines. In-field and real time auditing and monitoring of fertilizer application operations is done by Lead Farmers, BCP agricultural staff and front line Ministry of Agriculture camp and block extension staff. Possible adverse effects from direct contact are well known, and are normally restricted to minor irritations to the exposed area. First aid recommendations – usually washing out the exposed area normally resolves the issue. Neither product is classified as a carcinogen by NTP or IARC. Neither product is considered to be an explosion and or fire hazard. Both products have widely circulated Material Safety Data Sheets should additional information be required.
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5.2 Offsite Biodiversity Impacts

5.2.1 Negative Offsite Biodiversity Impacts (B3.1) and Mitigation Actions (B3.2)

Table 33. Negative Offsite Biodiversity Impacts and Mitigation Actions

Negative Offsite Impact	Mitigation Measure(s)
Activity leakage	As explained in section 5.1.2, the REDD+ activities are not expected to cause negative impacts on biodiversity inside and outside the project zone. Activity leakage affecting biodiversity through habitat degradation/loss is minimal as demonstrated by the monitoring of leakage plots (see VCS Monitoring Reports in Appendix B). The mitigation measures implemented to reduce the potential negative impact of project activities on community members previously involved in charcoal production and poaching (see section 4.3.2 of the PD) are also limiting negative biodiversity impacts. Furthermore, the presence of conservation organizations, hunting operators and the social enterprise COMACO active in the project area, are

Negative Offsite Impact	Mitigation Measure(s)
	contributing to the reduction of activity leakage affecting biodiversity, either through repressive or alternative measures⁴⁴. BCP-supported scouts are patrolling beyond the boundaries of the REDD+ zones (as shown in the SMART reports in Appendix G) and community impacts are taking place over the whole chiefdoms regardless of where the REDD+ zones boundaries fall. These activities are preventing, to a certain extent, biodiversity leakage from taking place.

5.2.2 Net Offsite Biodiversity Benefits (B3.3)

There is no negative impact of project activities on biodiversity outside the project zone that has been identified. The project is specifically designed to produce biodiversity benefits over an extensive area. The securing of a wildlife corridor between National Parks through the maintenance of suitable habitats under improved land management, effective surveillance and law enforcement will restore in the short-medium term the function of GMAs as buffer zones⁴⁵. Prior to the LCFP, snaring in the GMAs was a major threat for wildlife including for the African wild dogs⁴⁶. In the Luangwa valley, this carnivores species habitat in GMAs intensively, with some packs utilising them more frequently than National Parks, presumably because of the lower density of lions which act as both a higher competitor and predator. The condition suitability in GMAs could, however, be negated by higher snaring mortality, resulting in an “attractive sink” for the African wild dog population.

In addition, the cooperative behaviour of the African wild dogs, while extremely advantageous in terms of hunting and breeding success, can become a limiting factor in heavily snared areas. The various measures taken under the LCFP to reduce the poaching pressure in the REDD+ zone, along with other organizations, is therefore significantly contributing to the long-term survival of Zambia's largest population of African wild dogs⁴⁷. The African wild dogs can be seen as an umbrella species, where its protection benefits the whole ecosystem, and considering that the Luangwa valley is also home of Zambia's largest populations of hippos, elephants, lions and leopards, as well as endemic species (Thornicroft's Giraffe and Cookson's wildebeest), BCP can confidently say that the project has a net positive impact at a scale that goes far beyond the boundaries of the project, nationally (large carnivores, elephants and hippos) and globally in the case of endemic species.

In addition to securing important wildlife corridors and enhancing the effectiveness of the protected areas network, the improvement in the management of GMAs for which the REDD+ activities are contributing will

⁴⁴ Lewis, D. and Wilkie, D.S., 2020. COMACO, from snares to plowshares: A conservation and human wellbeing success story. *Conservation Science and Practice*, 2(10), p.e263.

⁴⁵ Lindsey, P., Nyirenda, V., Barnes, J., Becker, M., Taylor, A. and Watson, F., 2013. The reasons why Zambian game management areas are not functioning as ecologically or economically productive buffer zones and what needs to change for them to fulfil that role. *Lusaka, Zambia: Wildlife Producers Association of Zambia*.

⁴⁶ Becker, M., McRobb, R., Watson, F., Droge, E., Kanyembo, B., Murdoch, J. and Kakumbi, C., 2013. Evaluating wire-snare poaching trends and the impacts of by-catch on elephants and large carnivores. *Biological conservation*, 158, pp.26-36.

⁴⁷ Zambian Carnivore Programme. Annual Report. 2018

restore the function of GMAs as both a reservoir and refuge for wildlife. This will therefore benefit the offsite biodiversity.

5.3 Biodiversity Impact Monitoring

5.3.1 Biodiversity Monitoring Plan (B4.1, B4.2, GL1.4, GL3.4)

Between 2016 and 2021, biodiversity monitoring was conducted annually in the private Game Ranch of Munyamadzi Game Reserve. In order to obtain the required permission from the Department of National Parks and Wildlife to conduct monitoring activities in open areas and GMAs a concept note was submitted in early 2021. The permit was issued in February 2022 which enabled the expansion of the monitoring plan to new sites (See Appendix D). The details of the monitoring plan are presented in the table below.

Table 34. Details of the biodiversity monitoring plan in the LCFP.

Study site	Management	Method	Start date	Frequency	Variables
All sites	GMAs Private Open areas - communal	Path of least resistance walk	2020	Daily	presence/absence
		Questionnaire	2022	One-off	presence/absence
		GPS Trackers	2022	Continuous	Vultures' movement
Munyamadzi	Private	Ground Transect	2016	Biannually	Abundance Occupancy
		Camera trap	2020	Annually	Occupancy
Shikabeta	GMA	Ground Transect	2022	Annually	Abundance Occupancy
Mwanya	GMA	Ground Transect	2022	Annually	Abundance Occupancy
		Camera trap	2022	Annually	Occupancy

Results of the monitoring plan are detailed in the annual and survey reports available in Appendix D. In summary, the monitoring plan confirmed the presence of most HCV 1 attributes in the project area and indicates that in Munyamadzi where regular monitoring took place since the start of the project, wildlife populations are mostly stable or increasing.

5.3.2 Biodiversity Monitoring Plan Dissemination (B4.3)

Results of the biodiversity monitoring plan are published once a year on several platforms for general public use. First, a complete annual report is produced during the first quarter of the calendar year by the parties involved in the monitoring plan: Lion Landscapes and BCP (see Appendix D). Upon approval, each party is then able to share the results with other stakeholders and to post it on their websites. In the case of BCP, results are made available on the Impact Monitoring Website IM V2⁴⁸.

⁴⁸ <https://www.bcpimpactmonitoring.com/>

Summaries of the monitoring plan and results are shared with the local communities in the local language, in the form of handouts during community engagement meetings (Appendix J).

5.4 Optional Criterion: Exceptional Biodiversity Benefits

5.4.1 Trigger Species Population Trends (GL3.3)

Table 35. Trigger Species Population Trends: *Lycaon pictus* - African wild dogs (Endangered)

Trigger Species	<i>Lycaon pictus</i> - African wild dogs (Endangered)
With-project Scenario	The principal threat to African Wild Dogs is habitat fragmentation, which causes increased contact with people and domestic animals, resulting in human-wildlife conflict and transmission of infectious disease.⁴⁹ Another major threat, as observed in the Luangwa valley, is snaring by-catch⁵⁰. The LCFP tackles both threats through curbing deforestation and deploying surveillance and law-enforcement patrols. Recent documentation by the Zambian Carnivore Programme of wilddogs movements are demonstrating the importance of the LCFP as dispersing corridors for this species (Figure 3). The ongoing long-term monitoring of African wild dog populations in the Luangwa valley carried out by the Zambian Carnivore Programme is demonstrating a viable and stable population with around 200 individuals⁵¹. Likewise, the population of african wild dogs in the Lower Luangwa, mostly roaming the private Game Ranches, appears to have remained stable as suggested by the Occupancy analysis conducted in Munyamadzi and Kazumba (see Biodiversity Monitoring Annual Report 2021 in Appendix D).

Table 36. Trigger Species Population Trends: *Panthera leo* - Lion (Vulnerable)

Trigger Species	<i>Panthera leo</i> - Lion (Vulnerable)
With-project Scenario	The main threats to Lions are indiscriminate killing, primarily as a result of retaliatory or pre-emptive killing to protect human life and livestock, and prey base depletion resulting from unsustainable habitat loss and poaching levels⁵². Furthermore, habitat loss results in subpopulations of lions becoming small and isolated which makes them extremely sensitive to local extinction due to stochastic events. The LCFP is addressing all of these threats through the various activities designed to maintain wildlife habitats and to restore healthy wildlife populations. These activities are directly targeting the agents and drivers of deforestation, which are often similar to those of defaunation. The sustainable

⁴⁹ Woodroffe, R. & Sillero-Zubiri, C. 2020. *Lycaon pictus* (amended version of 2012 assessment). *The IUCN Red List of Threatened Species* 2020. <https://dx.doi.org/10.2305/IUCN.UK.2020-1.RLTS.T12436A166502262.en>.

⁵⁰ Becker, M., McRobb, R., Watson, F., Droge, E., Kanyembo, B., Murdoch, J. and Kakumbi, C., 2013. Evaluating wire-snare poaching trends and the impacts of by-catch on elephants and large carnivores. *Biological conservation*, 158, pp.26-36.

⁵¹ Zambian Carnivore Programme. Annual report 2020.

⁵² Bauer, H., Packer, C., Funston, P.F., Henschel, P. & Nowell, K. 2016. *Panthera leo* (errata version published in 2017). *The IUCN Red List of Threatened Species* 2016. <https://dx.doi.org/10.2305/IUCN.UK.2016-3.RLTS.T15951A107265605.en>

	opportunities offered by the LCFP to poachers, charcoal producers, and the overall community, as well as the effective deployment of surveillance and law enforcement patrols are significantly enhancing the impact of the repressive actions undertaken by governmental and non-governmental organizations. The effectiveness of such collaborative efforts to reduce the human impact on the wildlife population was celebrated in 2020 which represented the first year in decades where no snared lions were reported in the Luangwa valley⁵³. It is also expected that the conservation measures implemented in the project area under the LCFP, resulting in a better management of the GMAs will, in the medium-term, increase lion prey abundances. BCP is developing a Human Wildlife Interaction programme as more human wildlife conflicts are anticipated with the return of wildlife. Through a participatory approach, the programme aims to find solutions to avoid and mitigate any potential conflicts including with lions, therefore avoiding retaliatory killing. The long-term ongoing monitoring of lion populations in the Luangwa valley carried out by the Zambian Carnivore Programme is demonstrating a viable and stable population with around 200 individuals, which is the most important population in Zambia⁵⁴. The population of lions in the Shikabeta REDD+ zone, in the Luano valley, seems to be experiencing a steep recovery as suggested by local stakeholders and confirmed empirically during the transect survey in 2022. Likewise, the population of lions in the lower Luangwa appears to be increasing as suggested by the Occupancy analysis conducted in Munyamadzi and Kazumba (see Biodiversity Monitoring Annual Report 2021 in Appendix D).
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Table 37. Trigger Species Population Trends: *Panthera pardus* - Leopard (Vulnerable)

Trigger Species	<i>Panthera pardus</i> - Leopard (vulnerable)
With-project Scenario	The principal threat to leopards in the Luangwa valley appears to be the depletion of prey⁵⁵. In the Lupande GMA, where few LCFP instances take place, it is estimated that leopards occur at a mean density of 10.5 leopards per 100 km², as opposed to 18.7 leopards per 100 km² in the South Luangwa National Park where prey abundance is higher. Similarly to the other two large carnivore species listed above, the with-project scenario is expected to benefit the population of leopards through the maintenance and gradual increase of prey abundance through habitat loss and poaching mitigation activities across vast areas of wildlife corridors.

⁵³ Conservation South Luangwa. Annual Report 2020.

⁵⁴ Zambian Carnivore Programme. Annual report 2020.

⁵⁵ Rosenblatt, E., Creel, S., Becker, M.S., Merkle, J., Mwape, H., Schuette, P. and Simpamba, T., 2016. Effects of a protection gradient on carnivore density and survival: an example with leopards in the Luangwa valley, Zambia. Ecology and Evolution, 6(11), pp.3772-3785.

Table 38. Trigger Species Population Trends: *Loxodonta africana* - African elephant (Endangered)

Trigger Species	<i>Loxodonta africana</i> - African elephant (Endangered)
With-project Scenario	Ivory poaching was a major cause of population decline of elephants in the 1970s and 1980s. The population of elephants in the Luangwa valley is thought to have declined from an initial 100,000 individuals to less than 20,000. After this period, the population began to recover until a new surge of ivory poaching began in 2006. For the first time in 25 years, the global population of African elephants is again experiencing a decline⁵⁶. However, the last national elephant carcass count surveys in Zambia suggest that poaching of elephants in the Luangwa valley is low. Data on wildlife crime from DNPW indicated that 42 elephants were killed between 2016 and 2018⁵⁷. Elephants are also subject to snare injuries and retaliatory killing. As explained above, the project activities are designed to reduce threats to wildlife populations through reduced poaching, human-wildlife interaction mitigation and habitat loss prevention. The combined effort of the various organizations resulted in a reduction of elephant mortality by 66% between 2018 and 2020 in the northern part of the LCFP, notably thanks to a three-fold reduction in the number of poached elephants⁵⁸.

Table 39. Trigger Species Population Trends: *Hippopotamus amphibius* - Hippopotamus (Vulnerable)

Trigger Species	<i>Hippopotamus amphibius</i> - Hippopotamus (Vulnerable)
With-project Scenario	Zambia has historically been a stronghold for the hippopotamus population and recent country-wide estimates confirm this status. The Luangwa River is home by far to the largest population⁵⁹ with as many as 48 hippos per km of river stretch in certain areas⁶⁰. The population of the Luangwa River presents fluctuations naturally occurring in sites where the carrying capacity is already reached. Generally, the hippopotamus population is stable across Zambia, however, there are increasing concerns for future viability of these populations, particularly outside protected areas where human settlements and fishing activities increased, leading to increased Human-Hippo Conflicts from crop raiding or hippo attacks. Other threats include habitat loss or degradation through agricultural development and farming, and illegal and unregulated hunting for meat and ivory (found in the canine teeth). Human encroachment on hippo's habitat from land conversion and fishing results in an increased number of interactions, which could be further exacerbated during episodes of drought. Poaching is also a major threat, mostly in areas of civil unrest but also

⁵⁶ African Elephant Database (AED). <http://www.elephantdatabase.org>

⁵⁷ Lewis, D. and Wilkie, D.S., 2020. COMACO, from snares to plowshares: A conservation and human wellbeing success story. *Conservation Science and Practice*, 2(10), p.e263

⁵⁸ Conservation South Luangwa. Annual Report 2020.

⁵⁹ Wilbrod C and Milanzi J, 2010. Population status of the hippopotamus in Zambia. *African Journal of Ecology* 49, 130-132.

⁶⁰ Chomba, C., Simpamba, T., & Nyirenda, V. (2021). The Population Dynamics of the Luangwa (Zambia) Common Hippopotamus (*Hippopotamus amphibius*) during the Period 1952 – 2015. *International Research in Environment, Geography and Earth Science* Vol. 9, 74–91.

	in stable countries like Zambia where poaching was suggested to be the main factor explaining the low number of hippos present within the Zambezi River system. The LCFP is addressing both the habitat loss or degradation and the poaching pressure. Hippopotamus are nocturnal grazers consuming up to 50 kg of grass daily usually no further than 5 km from the water body. In the Luangwa valley where the hippopotamus population is high, large tracts of intact or conflict-free land are required to maintain the carrying capacity and the LCFP is significantly contributing to it by improving the management of up to 230,000 ha of grazing habitats (ie. area of LCFP within 5-km of the Luangwa and main tributaries).
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Table 40. Trigger Species Population Trends: Vultures (3 species Critically Endangered and 1 species Endangered)

Trigger Species	<i>Gyps africanus</i> - White-backed vulture (Critically Endangered) <i>Necrosyrtes monachus</i> - Hooded vulture (Critically Endangered) <i>Torgos tracheliotos</i> - Lappet-faced vulture (Endangered) <i>Trygoniceps occipitalis</i> - White-headed vulture (Critically Endangered)
With-project Scenario	With current rate of decline, it is expected that the population of the four species of vulture found in LCFP would collapse by 80-96% over the course of three generations⁶¹. The major threats listed are poisoning (intentional or unintentional), decline of food availability and habitat loss and degradation⁶². The LCFP is tackling all three threats. As explained earlier, the improved management of a large tract of forest through REDD+ activities and the resulting reduced rate of habitat loss and degradation is expected to benefit the wildlife populations and therefore to increase vultures' food availability. Furthermore, the causes and outcomes of poisoning are taken into consideration through a partnership with BirdWatch Zambia. The monitoring of the three tagged vultures in the Luangwa valley are showing movements from the north of the South Luangwa National Parks at the north of the LCFP all the way down the Luano valley in the Shikabeta chiefdom (Figure 3). The LCFP is therefore contributing to the maintenance of safe zones for vultures along an important migratory route.

5.4.2 Effectiveness of Threat Reduction Actions (GL3.4)

The main threats are habitat loss/fragmentation and poaching, both leading to a reduction of prey population abundance. In the case of carnivores, snare-induced injuries and mortality, conflictual interactions between humans and wildlife (crop raiding and livestock predation) could lead to retaliatory killing, and genetic bottleneck. The LCFP is specifically designed to address the loss of forest habitat and implement activities

⁶¹ Ogada, D., Shaw, P., Beyers, R.L., Buij, R., Murn, C., Thiollay, J.M., Beale, C.M., Holdo, R.M., Pomeroy, D., Baker, N. and Krüger, S.C., 2016. Another continental vulture crisis: Africa's vultures collapsing toward extinction. Conservation Letters, 9(2), pp.89-97.

⁶² Botha, A. J., Andevski, J., Bowden, C. G. R., Gudka, M., Safford, R. J., Tavares, J. and Williams, N. P. (2017). Multi-species Action Plan to Conserve African-Eurasian Vultures. CMS Raptors MOU Technical Publication No. 5. CMS Technical Series No. 35.

to mitigate Human Wildlife Conflicts. Many other organizations are also conducting anti-poaching and habitat protection activities. Most notably, Conservation South Luangwa, the Zambian Carnivores Programme and Wildlife Crime Prevention are doing immense work which has resulted in very positive trends of poaching reduction (see Annual Reports on their respective websites). While these organizations mainly focus on protecting wildlife through research and monitoring (ZCP), HWC mitigation (ZCP, CSL) and law enforcement (WCP, CSL), BCP is also focusing on protecting the wildlife habitat, ensuring connectivity between populations and food resource availability. As a result of all stakeholders combined efforts, the number of snared lions and African wild dogs has reduced in recent years, as did the number of elephants killed. The rate of forest loss also encompassed a reduction as shown by our remote sensing analysis, aerial surveys and leakage monitoring. These trends are very encouraging and are supporting the effectiveness of the measures taken to prevent further habitat and wildlife population loss, including LCFP trigger species.

6 ADDITIONAL PROJECT IMPLEMENTATION INFORMATION

Not applicable. All the relevant information is presented in the current report and the appendices available for the VVB.

7 ADDITIONAL PROJECT IMPACT INFORMATION

Not applicable. All the relevant information is presented in the current report and the appendices available for the VVB.

8 LIST OF ANNEXES

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	Summary sheet beekeeping trainings	XLS	1.2; 4.1.1
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	LCFP_ER_Calc_M4_Second_Activity_Instance	XLS	3.1.1
	LCFP_ER_Calc_M4_First_Activity_Instance	XLS	3.1.1
	LCFP_ER_Calc_M5_Third_Activity_Instance	XLS	3.1.1
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	Professor Chudimayo Report	PDF	2.1.2
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